

Trailblazing Tourism: A Data Science Look at Global Hiking Destinations

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1. Introduction



International travel and tourism is one of the largest global industries. According to the World Travel and Tourism Council, it represented 10% of the global economy in its contribution in 2024 (*Travel & Tourism Economic Impact Research (EIR)*, n.d.). Within travel, hiking is a rapidly growing segment. Around the world, places such as The Matterhorn Circuit of Switzerland, The Dolomites' Alta of Italy, and the Volcanoes of Guatemala, bring millions of visitors annually (Redd, 2025). These hiking decisions are formed by unique features that vary from trail to trail, such as trail difficulty, climate, accessibility, and cultural history. Visitor data allows tourism councils to better manage the impact it has on the environment and how they could create safe destinations for both beginner and more advanced travelers.

Hiking within the tourism segment affects multiple groups of people. The revenue it brings in can affect local governments and their policies. The environment is affected by heavy foot traffic and the impact it has on our ecosystems that many people strive to preserve. We also have to consider how climate change influences travel behavior, such as wildfires in the western United States, or melting glaciers in places such as the Himalayas. It is very important that we are able to anticipate and plan for future challenges that climate change will continue to bring. By studying data related to trail features, international travel number, and visitor reviews, we can

better understand what destinations are suitable for certain people (such as families or beginners) and how factors can affect visitation patterns.

I chose this topic for a few reasons. Hiking is extremely important to me. It is a way that I can connect with nature and myself, while reducing stress and exploring new beautiful terrains. I grew up in Moab, Utah and my parents moved us there at a young age. This is where my love of hiking originated from. Moab also has a very large tourism sector and people travel from all over the world to experience the unique rock formations, so I am very familiar with the outdoor tourism industry. In addition, I love to travel and though I have only been international once, it was an amazing experience that I hope to repeat many times. When brainstorming for this project, I wanted to get more insight of what actually drives people to visit certain places over others, in terms of popular hiking destinations. In summary, exploring these questions that affect tourism can combine my curiosity with my passion for the outdoors.

As global travel continues to grow, new patterns and motivations are shaping how and where people choose to explore. One key curiosity driving this project is whether certain countries naturally attract beginner hikers while others draw more adventure seekers with an interest in high elevation treks. Another guiding question is whether environmental factors, such as wildfire seasons or weather patterns, influence tourist behaviors. In addition, reviews and traveler sentiment may reveal insight into what people value most with trails. To sum this up, the main goal of this project is to have a better understanding of what drives people to travel for hiking and whether these patterns align with climate, difficulty level, and characteristics of destinations.

Questions to drive research

1. Which countries attract the most hiking focused international tourists?
 - a. Initial Idea: Compare total number of trails and trail reviews by country. We can do this by referencing the number of arrivals per country (a good source is the World Bank). Methods/models could include descriptive analysis through counts in tables and bar charts. We would also be using predictive analysis by using historical data.
2. Which countries attract the most difficult/advanced hikes?
 - a. Initial Idea: Filter trails with a “hard” or “advanced” difficulty and see which countries or regions have a higher amount compared to others. We can also look at elevation gain and trail length to determine how “hard” a trail is. Descriptive analysis can be used by clustering and filtering countries based on their amount of “hard” hikes.
3. Which countries attract the most beginner/family-friendly hikes?
 - a. Initial Idea: Similar to above, filtering trails with a “easy” difficulty to see which countries have a higher amount compared to others. We can also look at less elevation gain and shorter trail length to determine if a trail can be considered a beginner level. Descriptive analysis can be used by clustering and filtering countries based on their amount of “easy” hikes.
4. Can we predict the popularity of hiking decisions based on their trail features (such as trail length)?
 - a. Initial Idea: Use the amount of reviews a trail has, as well as their popularity/star rating. We can predict by using the trail length, elevation, and difficulty. The method would be predictive analysis, as we are using it to draw data patterns from. A decision tree might be useful in this scenario but I would need to familiarize myself more with this method.
5. Do tourism flows correlate with climate patterns (temperature, wildfire season, rainy season, etc.)?
 - a. Initial Idea: Merge data such as the number of international tourists per country (from the World Bank arrivals source), as well as climate patterns for the same country and the same year. I found a source, the World Bank Climate Change Knowledge Portal, that has data for annual temperature anomalies per country and per year. Combining these data sets by country/year and using exploratory analysis, we can compare drought or wildfire years to see if arrivals dropped during that time. This would be using diagnostic analysis, as we are looking for correlations and finding patterns.
6. Are there clear clusters of countries that are more hiking destinations over more leisure tourism?
 - a. Initial idea: Compare amount of trails bs. Overall tourist arrivals per country with the World Bank arrivals source. Countries with higher trail to tourist ratios could be considered “hiking destinations”. This could be an unsupervised method of clustering using descriptive analysis and the use of K-means clustering.

7. What words appear more commonly in reviews of highly rated destinations?
 - a. Initial idea: Analyze text reviews from Alltrails data set. We can filter by focusing on the trails with a star rating of 4 or greater and look for repeated words within those reviews. This is a form of descriptive analysis as we are summarizing the reviews associated with 4 stars or greater and using mining of the qualitative data.
8. What words appear more commonly in reviews of lower rated hiking destinations?
 - a. Initial idea: Similar to above, analyze text reviews from Alltrails data set. We can filter by focusing on the trails with a star rating of 2 or lower and look for repeated words within those reviews. This is a form of descriptive analysis as we are summarizing the reviews associated with 2 stars or less and using mining of the qualitative data.
9. Are there certain continents that have a more hiking/adventure tourism sector?
 - a. Initial idea: Group the trail and review data by continent (with the Alltrails data set), and focus on the average difficulty of the hike as well as the number of reviews. Descriptive analysis is performed by comparing the patterns of trail difficulty vs international arrivals (World Bank source).
10. Can we forecast future hiking tourism growth in emerging destinations?
 - a. Initial idea: Identify countries with relatively smaller but growing hiking sectors. Use historical tourism arrival data (World Bank) and the amount of trail availability in order to model the growth that has happened. In other words, we could filter the countries that have fewer trails and fewer reviews and focus on those. We use predictive analysis based on past observations over time and use a line graph in order to track this data for those countries. The x axis would be the time in years, while the y axis could be the number of international tourist arrivals.

Data Gathering

Direct Data Downloads

Alltrails Trek Data

This dataset contains hiking trail information collected from AllTrails, the leading hiking trail companion app. It includes trail names, trail images, geographical location, length, elevation gain, difficulty level, average ratings from users, as well as number of user reviews. The dataset is labeled with each column corresponding to the above variables. It also contains both qualitative and quantitative data. For instance, the hiking destination is in the form of qualitative data while the average rating is quantitative.

[Link to Keggale Data Set](#)

Trail_name	link_AllTrails	image	Difficulty	Average_rating	number_of_reviews	Location	Location	Length	A description					
754 unique values	755 unique values	755 unique values	Hard Moderate Other (124)	43% 42% 15%		(2) (3) Other (538)	19% 17% 64%	[null] Pune, Maharashtra... Other (496)	38% 3% 59%	[null] Hemis National Park Other (294)	62% 3% 35%	571 unique values	[null] This is one of the t... Other (812)	3% 1% 96%
Triund Trek Loop	https://www.alltrails.com/trail/india/himachal-pradesh/triund-trek-loop	https://www.alltrails.com/api/alltrails/v2/trails/10641033/p/hotos/?size=larger_wide&key=3p0t5s6b5g4...	Hard	4.6	(72)	Dharmshala, Himachal Pradesh, India		Length: 19.0 km	This is one of the top tourist destinations in the vicinity of Dharmshala, Himachal Pradesh. Spectac...					
Lamadugh Natural Valley Trek	https://www.alltrails.com/trail/india/himachal-pradesh/lamadugh-natural-valley-trek	https://www.alltrails.com/api/alltrails/v2/trails/10667708/p/hotos/?size=larger_wide&key=3p0t5s6b5g4...	Hard	4.3	(86)		Manali Wildlife Sanctuary	Length: 11.4 km	Lamadugh (Also known as Lamadug) is a popular full-day trek in Manali. The trail starts from Bhungr...					
Sinhagad Fort	https://www.alltrails.com/trail/india/maharashtra/sinhagad-fort	https://www.alltrails.com/api/alltrails/v2/trails/10611807/p/hotos/?size=larger_wide&key=3p0t5s6b5g4...	Hard	4.8	(227)	Velhe, Maharashtra, India		Length: 8.0 km	Sinhagad Fort is a fort that was built 2000 years ago, and references to this fort can be found in t...					
Blackberry Hill - Attukad Falls Loop	https://www.alltrails.com/trail/india/kerala/blackberry-hill-attukad-falls-loop	https://www.alltrails.com/api/alltrails/v2/trails/11136907/p/hotos/?size=larger_wide&key=3p0t5s6b5g4...	Moderate	4.6	(64)	Munnar, Kerala, India		Length: 9.3 kmEst. 3h 10m	This is a beautiful mountain hiking trail in Munnar. It goes through meadows, waterfalls, and the ma...					
Triund Hill Trek	https://www.alltrails.com/trail/india/himachal-pradesh/triund-hill-trek	https://www.alltrails.com/api/alltrails/v2/trails/11267924/p/hotos/?size=larger_wide&key=3p0t5s6b5g4...	Hard	4.6	(73)	Dharamsala, Himachal Pradesh, India		Length: 9.5 km	This is one of the top tourist destinations in the vicinity of Dharmshala, Himachal Pradesh. Spectac...					
Avalahalli Forest Loop	https://www.alltrails.com/trail/india/karnataka/avalahalli-forest-loop	https://www.alltrails.com/api/alltrails/v2/trails/10586259/p/hotos/?size=larger_wide&key=3p0t5s6b5g4...	Easy	4.6	(70)		Avalahalli State Forest	Length: 6.8 kmEst. 1h 38m	Avalahalli State Forest is known for its many mountain biking trails and this is one of the many. Al...					
Skandagiri Hills (Kalawara Betta)	https://www.alltrails.com/trail/india/karnataka/skandagiri-hills	https://www.alltrails.com/api/alltrails/v2/trails/10747372/p/hotos/?size=larger_wide&key=3p0t5s6b5g4...	Hard	4.5	(47)	Chikkaballapur, Karnataka, India		Length: 6.0 km	Skandagiri, also known as Kalawara Durga, is an ancient mountain fortress located approximately 70 k...					
Ram Jhula - Neelkanth Mahadev Trail	https://www.alltrails.com/trail/india/uttarakhand/ram-jhula-neelkanth-mahadev-trail	https://www.alltrails.com/api/alltrails/v2/trails/11012568/p/hotos/?size=larger_wide&key=3p0t5s6b5g4...	Hard	4.5	(38)		Rajaji National Park	Length: 8.2 kmEst. 3h 13m	Neelkanth Mahadev is a Temple in Rishikesh that is revered by tourists from all over the world and ...					
Kuniasauri Temple	httos://www.alltrail	httos://www.alltrail	Hard	4.8	(41)	Narendra Naagar.		Lenoeth: 22.5 km	A beautiful trail					

Alltrails (Gathered) Data

After cleaning the data from my Alltrails Trek Data, I discovered that after I used an API to find the locations of entries that only states names (such as sanctuaries), that the entire data set only contained trails from India. After further research, I found that Alltrails is very strict on providing datasets or allowing data scraping. In order to retrieve the information that was crucial to my project, I manually went through the Alltrails website and gathered my own data for over 50 countries.

1	Country	TrailCount	HardPercent	EasyPercent	AvgLen_mi	AvgElevGain_ft	AvgRating	ReviewCount
2	Australia	18,729	0.2	0.1	5.21	1058.5	4.2	1253359
3	Austria	8,419	0.3	0.05	5.66	1351.9	4.3	80138
4	Albania	59	0.7	0.05	6.93	2562.1	4.3	5975
5	Algeria	4	0.5	0.25	19.7	1140.5	4.4	23
6	Belize	36	0.25	0.15	2.81	530	4.5	956
7	Brazil	9,527	0.45	0.1	4.86	1748.5	4.5	27756
8	Canada	29,814	0.42	0.1	4.925	1710.05	4.2	4085482
9	Chile	2,280	0.45	0.15	10.02	2449.2	4.4	39530
10	China	548	0.75	0.1	7.48	29280	4.3	7924
11	Colombia	2,047	0.55	0.05	6.05	1681.9	4.5	13714
12	Costa Rica	1,589	0.05	0.15	3.01	624	4.4	30115
13	Croatia	4,192	0.2	0.2	4.99	917.7	4.3	46101
14	Cuba	21	0.25	0.3	8.8	1169.5	4.6	152
15	Denmark	9,079	0.05	0.1	8.42	787.2	4.1	191757
16	Dominican Repu	81	0.4333333333	0.1	11.04	3062.9	4.5	1903
17	Ecuador	227	0.6	0.1	7.71	5801.6	4.4	9330
18	Estonia	262	0	0.35	4.84	422.1	4.3	3309
19	Fiji	9	0.5555555556	0.1111111111	12.54444444	11272.4	4.6	255
20	Finland	5,561	0.2	0.1	4.48	701	4.2	18772
21	France	39,816	0.34	0	5.3025	1507.65	4.2	1154426
22	Georgia	208	0.7	0	10.02	3312.2	4.4	8691
23	Germany	37,885	0.15	0.25	4.06	1010.8	4.1	282913
24	Greece	3,787	0.3	0.05	4.52	1598.7	4.1	83454
25	Honduras	77	0.45	0.15	4.59	1784.7	4.5	850
26	Hungary	3,189	0.1	0.15	5.94	1163	4.3	16780
27	Iceland	652	0.2	0.4	5.07	1159.3	4.4	90741
28	Japan	5,833	0.48	0.04	5.545	1840.25	4.2	62140
29	Kazakhstan	25	0.65	0.05	6.515	2224.8	4.5	734
30	Kenya	20	0.45	0.35	8.95	3220.6	4.6	1146
31	Liechtenstein	315	0.5	0.1	5.79	1585.8	4.2	1919
32	Madagascar	23	0.4	0.15	15.93	3100.8	4.6	121
33	Malaysia	4,986	0.25	0	2.53	1090.1	4.3	194055
34	Mongolia	1	1	0	7	2572	4.5	52
35	Montenegro	80	0.7	0.05	7.03	2318.7	4.4	6113
36	Morocco	24	0.15	0.3	6.23	2342.3	4.3	2470
37	Namibia	33	0.05	0.35	5.02	1026.6	4.5	174
38	Nepal	67	0.9	0.05	26.6	12385.9	4.4	7691
39	New Zealand	5,627	0.6	0.05	6.41	2504.1	4.3	358492
40	Norway	9,124	0.50	0.05	5.94	2128.7	4.2	168672
41	Peru	2,012	0.9	0	6.36	2195.3	4.5	26877
42	Philippines	1,891	0.6	0.15	3.86	1274.2	4.2	5085
43	Portugal	9,433	0.2	0.1	5.08	1845.5	4.1	258188
44	Romania	1,175	0.6	0.05	5.86	2367.4	4.4	7485
45	Spain	23,696	0.55	0	6.3	1715.5	4.1	455144
46	Slovenia	657	0.25	0.20	4.04	967.1	4.3	24756
47	Sweden	8,283	0	0.15	5.46	718.7	4.1	112338

International Tourism- Number of Arrivals

This dataset from the World Bank gives the number of tourist arrivals annually by country. It includes country name, country code, and years ranging from 1960-2020. The dataset is labeled with each column corresponding to the mentioned variables. Though we will focus on cleaning data in further modules, I did remove the columns of years prior to 1995, because they didn't actually contain any data so it seemed pointless to have it. This dataset contains both qualitative and quantitative data. This data will be useful in determining how popular tourist destinations correlate to popular hiking trails.

[Link to Data Set](#)

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
Data Source	World Development Indicators														
Last Updated Date	2025-10-07														
Country Name	Country Code	Indicator Name	Indicator Code	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006
Aruba	ABW	International tourist arrivals	ST.INT.ARVL	912000	957000	947000	906000	972000	1211000	1178000	1225000	1184000	1304000	1286000	128500
Africa Eastern and Southern	AFE	International tourist arrivals	ST.INT.ARVL	11583544.71	13088654.2	13456245.97	14403851.83	15309377.84	15353177.3	15854696.29	17383375.42	17844385.28	18745951.16	19917566.37	22650320.8
Afghanistan	AFG	International tourist arrivals	ST.INT.ARVL												
Africa Western and Central	AFW	International tourist arrivals	ST.INT.ARVL	2670705.934	3027134.733	3243144.347	3422652.337	3897975.031	4162850.003	4615887.342	4697120.11	5013049.71	5839376.585	6320831.909	6955607.10
Angola	AGO	International tourist arrivals	ST.INT.ARVL	9000	21000	45000	52000	45000	51000	67000	91000	107000	194000	210000	121000
Albania	ALB	International tourist arrivals	ST.INT.ARVL	304000	287000	119000	184000	371000	317000	354000	470000	557000	645000	748000	937000
Andorra	AND	International tourist arrivals	ST.INT.ARVL					9422000	10991000	11351000	11507000	11601000	11668000	11049000	10737000
Arab World	ARB	International tourist arrivals	ST.INT.ARVL										80985285.3	82979842.63	89387603.6
United Arab Emirates	ARE	International tourist arrivals	ST.INT.ARVL												
Argentina	ARG	International tourist arrivals	ST.INT.ARVL	2289000	2614000	2764000	3012000	2898000	2909000	2620000	2820000	2995000	3457000	3823000	4173000
Armenia	ARM	International tourist arrivals	ST.INT.ARVL	12000	13000	23000	32000	41000	45000	123000	162000	206000	263000	319000	382000
American Samoa	ASM	International tourist arrivals	ST.INT.ARVL	34000	35000	26000	36000	41000	44000	36000			24500	25299.9992	
Antigua and Barbuda	ATG	International tourist arrivals	ST.INT.ARVL	447000	498000	526000	570000	560000	634000	624000	528000	623000	769000	712000	838000
Australia	AUS	International tourist arrivals	ST.INT.ARVL	3726000	4165000	4318000	4167000	4459000	4931000	4856000	4841000	4746000	5215000	5499000	5532000
Austria	AUT	International tourist arrivals	ST.INT.ARVL	17173000	17090000	16647000	17352000	17467000	17982000	18180000	18611000	19078000	19374000	19952000	20269000
Azerbaijan	AZE	International tourist arrivals	ST.INT.ARVL	93000	210000	306000	483000	602000	681000	767000	834000	1038000	1279000	1177000	1194000
Burundi	BDI	International tourist arrivals	ST.INT.ARVL	34000	27000	11000	15000	26000	29000	36000	74000	74000	133000	148000	214000
Belgium	BEL	International tourist arrivals	ST.INT.ARVL	5560000	5829000	6037000	6179000	6369000	6457000	6452000	6720000	6690000	6710000	6747000	6995000
Benin	BEN	International tourist arrivals	ST.INT.ARVL	580000	516000	541000	575000	941000	1068000	1155000	853000	850000	845000	960000	975000
Burkina Faso	BFA	International tourist arrivals	ST.INT.ARVL	124000	131000	138000	160000	117000	126000	128000	150000	163000	222000	245000	264000
Bangladesh	BGD	International tourist arrivals	ST.INT.ARVL	156000	166000	182000	172000	173000	199000	207000	207000	245000	271000	208000	200000
Bulgaria	BGR	International tourist arrivals	ST.INT.ARVL	8005000	6811000	7543000	5240000	5056000	4922000	5104000	5563000	6241000	6982000	7282000	7499000
Bahrain	BHR	International tourist arrivals	ST.INT.ARVL	2311000	1988000	2600000	2890000	3280000	3869000	4388000	4831000	4844000	5667000	6313000	7289000
Bahamas, The	BHS	International tourist arrivals	ST.INT.ARVL	3239000	3416000	3446000	3348000	3648000	4204000	4183000	4406000	4594000	5004000	4779000	4731000
Bosnia and Herzegovina	BIH	International tourist arrivals	ST.INT.ARVL			115000	148000	147000	171000	139000	160000	165000	190000	217000	256000
Belarus	BLR	International tourist arrivals	ST.INT.ARVL						2030000	1852000	1907000	1729000	1816000	4738000	5276000
Belize	BLZ	International tourist arrivals	ST.INT.ARVL	342000	349000	305000	300000	340000	374000	385000	694000	999000	1329000	1037000	903000
Bermuda	BMU	International tourist arrivals	ST.INT.ARVL	557000	570000	562000	558000	549000	539000	458000	484000	483000	478000	517000	635000
Bolivia	BOL	International tourist arrivals	ST.INT.ARVL	284000	313000	356000	387000	342000	319000	316000	334000	427000	480000	524000	521000
Brazil	BRA	International tourist arrivals	ST.INT.ARVL	1991000	2666000	2850000	4818000	5107000	5313000	4773000	3785000	4133000	4794000	5358000	5017000
Barbados	BRB	International tourist arrivals	ST.INT.ARVL	927000	957000	990000	1019000	947000	1078000	1035000	1021000	1090000	1273000	1111000	1102000
Brunei Darussalam	BRN	International tourist arrivals	ST.INT.ARVL	498000	837000	643000	964000	967000	984000	840000	891000	944000		815000	836000
Bhutan	BTN	International tourist arrivals	ST.INT.ARVL	4800.000191	5199.999809	5400.000095	6199.999809	7199.999809	7599.999905	6400.000095	5599.999905	6300.000191	9199.999809	13600.00038	17299.9992
Botswana	BWA	International tourist arrivals	ST.INT.ARVL	636000	656000	765000	940000	1039000	1306000	1451000	1485000	1592000	1727000	1684000	1642000
Central African Republic	CAF	International tourist arrivals	ST.INT.ARVL	26000	21000	17000	7500	10000	11199.99981	9899.999619	2900.000095	5699.999809	8199.999809	12000	13800.00011
Canada	CAN	International tourist arrivals	ST.INT.ARVL	41657000	43256000	45076000	48064000	49055000	48638000	47147000	44896000	38903000	38845000	36160000	33390000
Central Europe and the Baltics	CEB	International tourist arrivals	ST.INT.ARVL	180652644.9	199115966.3	203579244.	200151256.8	200062962	206946379.3	182538824.4	173510081.4	177955845.4	200553517.6	208136165.2	215062085.1
Switzerland	CHE	International tourist arrivals	ST.INT.ARVL	6946000	6730000	7039000	7185000	7154000	7821000	7455000	6868000	6530000		7229000	7863000
Channel Islands	CHI	International tourist arrivals	ST.INT.ARVL												
Chile	CHL	International tourist arrivals	ST.INT.ARVL												
China	CHN	International tourist arrivals	ST.INT.ARVL	46387000	51128000	57588000	63478000	72796000	83444000	89013000	97908000	91662000	109038000	120292000	124942000
Cote d'Ivoire	CIV	International tourist arrivals	ST.INT.ARVL	188000	237000	274000	301000								
Cameroon	CMR	International tourist arrivals	ST.INT.ARVL												451000
Congo, Democratic Republic of the	COD	International tourist arrivals	ST.INT.ARVL	35000	37000	30000	53000	80000	103000	55000		35000	36000	61000	55000
Congo, Republic of the	COG	International tourist arrivals	ST.INT.ARVL												
Croatia	HRV	International tourist arrivals	ST.INT.ARVL	4433000	3740000	3700000	4730000	7370000	7030000	7080000	6740000	6600000	8000000	8840000	9000000

Country Latitude and Longitude Coordinates

In the API mentioned below, I will need to feed it coordinates in order to receive the information that is required. I am trying to find historical weather temperatures by country while the API requires specific coordinates to a city. This data set that I found called 'World Educational Data', gives the country name and the latitude/longitude coordinates for the center of the country. It contains both qualitative and quantitative data. I went ahead and removed the data that I do not need. There is only three columns, but this is not one of my required datasets I am using.

[Link to Data Set](#)

	A	B	C
1	Countries and ar	Latitude	Longitude
2	Afghanistan	33.93911	67.709953
3	Albania	41.153332	20.168331
4	Algeria	28.033886	1.659626
5	Andorra	42.506285	1.521801
6	Angola	11.202692	17.873887
7	Anguilla	18.220554	63.068615
8	Antigua and Barb	17.060816	61.796428
9	Argentina	38.416097	63.616672
10	Armenia	40.069099	45.038189
11	Australia	25.274398	133.775136
12	Austria	47.516231	14.550072
13	Azerbaijan	40.143105	47.576927
14	The Bahamas	25.03428	77.39628
15	Bahrain	26.0667	50.5577
16	Bangladesh	23.684994	90.356331
17	Barbados	13.193887	59.543198
18	Belarus	53.709807	27.953389
19	Belgium	50.503887	4.469936
20	Belize	17.189877	88.49765
21	Benin	9.30769	2.315834
22	Bhutan	27.514162	90.433601
23	Bolivia	16.290154	63.588653
24	Bosnia and Herz	43.915886	17.679076
25	Botswana	22.328474	24.684866
26	Brazil	14.235004	51.92528
27	British Virgin Isla	18.420695	64.639968
28	Brunei	4.535277	114.727669
29	Bulgaria	42.733883	25.48583
30	Burkina Faso	12.238333	1.561500

Data Gathering Using an API

Historical Forecast API

This API by Open Meteo gives a time series of historical weather data by using multiple data sources that provide weather data for specific regions. There are many different variables you can use in your query including: max temp/ min temp/, max wind speed, sunrise, sunset, daylight duration, precipitation sum, snowfall, etc (you can give a date range, so we can do it annually). The columns that I will be generating from this will include:

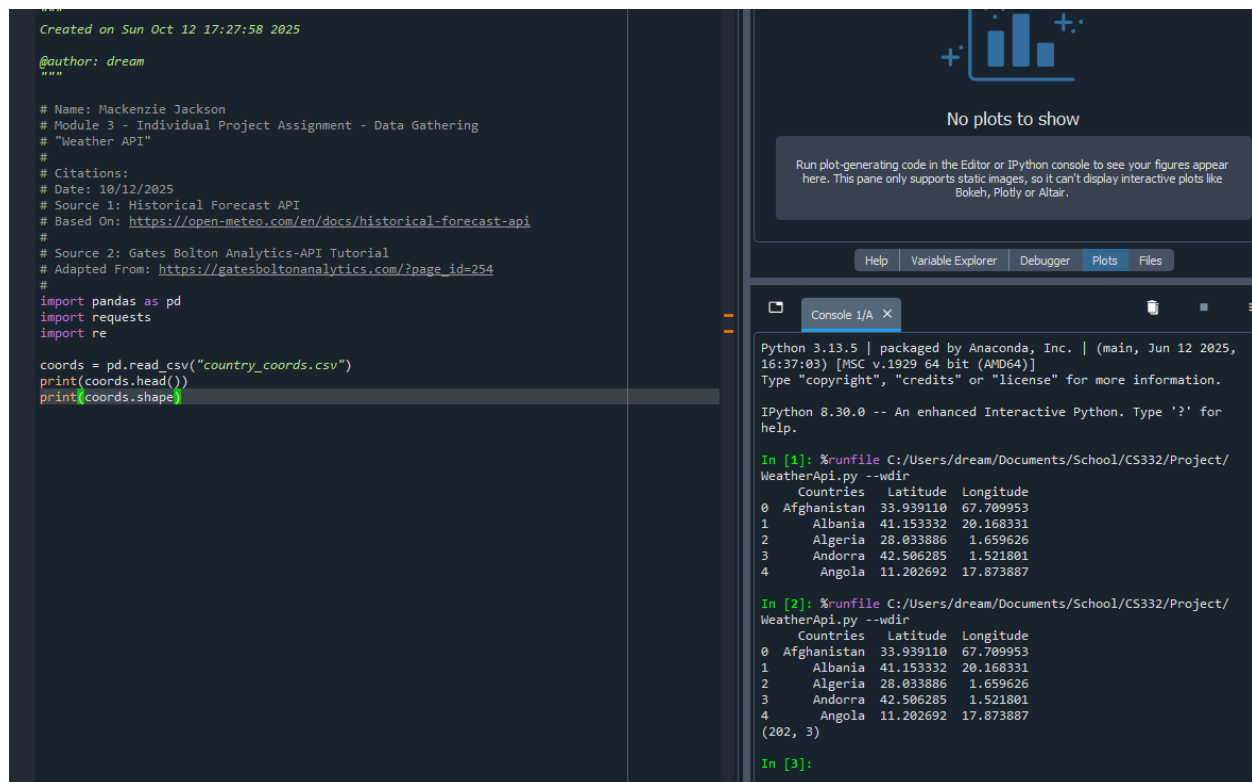
- Latitude
- Longitude
- Year
- Mean Temperature
- Max Temperature
- Min Temperature
- Precipitation Sum
- Sunshine Duration
- Radiation Sum

[Historical Forecast API link](#)

Example call for one country (Afghanistan):

https://archive-api.open-meteo.com/v1/archive?latitude=33.93911&longitude=67.709953&start_date=1990-01-01&end_date=1990-12-31&daily=temperature_2m_mean,temperature_2m_max,temperature_2m_min,precipitation_sum,sunshine_duration,shortwave_radiation_sum&timezone=auto&temperature_unit=fahrenheit&precipitation_unit=mm

1. The first thing I did was read in my country coordinates from the date file mentioned above. I printed the data head and shape, in order to see a snippet of the data as well as the number of columns/rows it contained. It appeared to load successfully.



- Next, I followed along with the doc that Open Meteo gave to set this API up. I had to install the necessary packages listed. They recommended setting it up with cached sessions and this retry logic. Luckily, the structure was pretty similar to the tutorial for the news API so I was able to understand what was going on and edit the parameters as I needed. Also, it was very easy to set up because you could check the boxes of attributes you needed and it generated everything for you. Though my goal is to get data for all the countries, I started by hard coding in variables for Afghanistan to make sure it was retrieving it correctly. I defined the endpoint (URL) of my API and set up the parameters that I will be using.

The screenshot displays the Spyder Python IDE interface. The main editor shows a Python script named `WeatherApi.py` with the following content:

```

1  # -*- coding: utf-8 -*-
2  """
3  Created on Sun Oct 12 17:27:58 2025
4
5  @author: dream
6  """
7
8  # Name: Mackenzie Jackson
9  # Module 3 - Individual Project Assignment - Data Gathering
10 # "Weather API"
11 #
12 # Citations:
13 # Date: 10/12/2025
14 # Source 1: Historical Forecast API
15 # Based On: https://open-meteo.com/en/docs/historical-forecast-api
16 #
17 # Source 2: Gates Bolton Analytics-API Tutorial
18 # Adapted From: https://gatesboltonanalytics.com/?page_id=254
19 #
20 import openmeteo_requests
21 import pandas as pd
22 import requests_cache
23 from retry_requests import retry
24 import numpy as np
25
26 coords = pd.read_csv("country_coords.csv")
27
28 # Setup cached session and retry logic
29 cache_session = requests_cache.CachedSession('.cache', expire_after=3600)
30 retry_session = retry(cache_session, retries=5, backoff_factor=0.2)
31 openmeteo = openmeteo_requests.Client(session=retry_session)
32
33 lat = 33.93911
34 lon = 67.709953
35 year = 2019
36
37 url = "https://historical-forecast-api.open-meteo.com/v1/forecast"
38 params = {
39     "latitude": lat,
40     "longitude": lon,
41     "start_date": f"{year}-01-01",
42     "end_date": f"{year}-12-31",
43     "daily": [
44         "temperature_2m_mean",
45         "temperature_2m_max",
46         "temperature_2m_min",
47         "precipitation_sum",
48         "sunshine_duration",
49         "shortwave_radiation_sum"
50     ],
51     "timezone": "auto",
52     "temperature_unit": "fahrenheit",
53     "precipitation_unit": "mm"
54 }
55 responses = openmeteo.weather_api(url, params=params)
56 response = responses[0]
57
58 print(f"Coordinates: {response.Latitude()}°N {response.Longitude()}°E")
59 print(f"Elevation: {response.Elevation()} m asl")
60

```

The right-hand pane shows the Variable Explorer with the following variables:

Name	Type	Size	Value
cache_session	session.CachedSession	1	CachedSession object of requests_cache.session module
coords	DataFrame	[202, 3]	Column names: Countries, Latitude, Longitude
lat	float	1	33.93911
lon	float	1	67.709953
openmeteo	Client.Client	1	Client object of openmeteo_requests.Client module
params	dict	8	{'latitude': 33.93911, 'longitude': 67.709953, 'start_date': '2019-01-01' ...
response	WeatherApiResponse.WeatherApiResponse	1	WeatherApiResponse object of openmeteo_sdk.WeatherApiResponse module
responses	list	1	[WeatherApiResponse]
retry_session	session.CachedSession	1	CachedSession object of requests_cache.session module
url	str	58	https://historical-forecast-api.open-meteo.com/v1/forecast
year	int	1	2019

The bottom pane shows the Console output:

```

Python 3.13.5 | packaged by Anaconda, Inc. | (main, Jun 12 2025, 16:37:03) [MSC v.1929 64 bit (AMD64)]
Type "copyright", "credits" or "license()" for more information.

In [1]: %runfile C:/Users/dream/Documents/School/CS332/Project/WeatherApi.py --wdir
Coordinates: 33.875°N 67.75°E
Elevation: 3777.0 m asl

In [2]:

```

- So now I want my API to retrieve information for each country based on the coordinates from my other CSV file and put that data in a new file. I will create a new list called results and then loop through my csv file. I also need to loop through a timespan of multiple years, but for now I want to get this down and then will later loop through each year I want. I then need to process the data into the pandas data frame. After I did this I ran into a problem where it was timing out because I suppose I was asking for too many requests from the API server by looping through all my countries. I was able to bypass this by putting a sleep of 2 seconds after every retrieval. It did take a bit to get all of the data, but I wasn't for sure how to bypass this if this is a stipulation with the API itself.

```
37 country = row["Country"]
38 lat = row["Latitude"]
39 lon = row["Longitude"]
40 year = 2019
41
42
43 url = "https://archive-api.open-meteo.com/v1/archive"
44 params = {
45     "latitude": lat,
46     "longitude": lon,
47     "start_date": f"{year}-01-01",
48     "end_date": f"{year}-12-31",
49     "daily": [
50         "temperature_2m_mean",
51         "temperature_2m_max",
52         "temperature_2m_min",
53         "precipitation_sum",
54         "sunshine_duration",
55         "shortwave_radiation_sum"
56     ],
57     "timezone": "auto",
58     "temperature_unit": "fahrenheit",
59     "precipitation_unit": "mm"
60 }
61 responses = openmeteo.weather_api(url, params=params)
62 response = responses[0]
63 daily = response.Daily()
64
65 daily_data = {
66     "date": pd.date_range(
67         start=pd.to_datetime(daily.Time(), unit="s", utc=True),
68         end=pd.to_datetime(daily.TimeEnd(), unit="s", utc=True),
69         freq=pd.Timedelta(seconds=daily.Interval()),
70         inclusive="Left"
71     ),
72     "country": country,
73     "temperature_mean_f": daily.Variables(0).ValuesAsNumpy(),
74     "temperature_max_f": daily.Variables(1).ValuesAsNumpy(),
75     "temperature_min_f": daily.Variables(2).ValuesAsNumpy(),
76     "precipitation_mm": daily.Variables(3).ValuesAsNumpy(),
77     "sunshine_duration_s": daily.Variables(4).ValuesAsNumpy(),
78     "shortwave_radiation_mj": daily.Variables(5).ValuesAsNumpy()
79 }
80
81 df = pd.DataFrame(daily_data)
82 results.append(df)
83 print(f"Retrieved data for {country}")
84 time.sleep(5)
85
86 if results:
87     final_df = pd.concat(results, ignore_index=True)
88     final_df.to_csv("climate_data_by_country.csv", index=False)
89     print("Saved as climate_data_by_country.csv")
```

Variable	Type	Count
cache_session	session.CachedSession	1
coords	DataFrame	[2]
country	str	8
daily	VariablesWithTime.VariablesWithTime	1
daily_data	dict	8
df	DataFrame	[3]
final_df	DataFrame	[7]
lat	float	1

Help Variable Explorer Debugger Plots Files

Console 2/A

- Retrieved data for Tonga
- Retrieved data for Trinidad and Tobago
- Retrieved data for Tunisia
- Retrieved data for Turkey
- Retrieved data for Turkmenistan
- Retrieved data for Turks and Caicos Islands
- Retrieved data for Tuvalu
- Retrieved data for Uganda
- Retrieved data for Ukraine
- Retrieved data for United Arab Emirates
- Retrieved data for United Kingdom
- Retrieved data for Tanzania
- Retrieved data for United States
- Retrieved data for Uruguay
- Retrieved data for Uzbekistan
- Retrieved data for Vanuatu
- Retrieved data for Venezuela
- Retrieved data for Vietnam
- Retrieved data for Yemen
- Retrieved data for Zambia
- Retrieved data for Zimbabwe

Saved as climate_data_by_country.csv

In [7]:

I ran into an issue here and realized that I was storing data for every day of every hour. That's a lot of data for just one year! What I needed to do was find the totals/averages of the year after retrieving the data but before storing it in a new file.

- For the next step, I need to implement averaging out the information that I actually need before storing it in a new file (as mentioned above), as well as loop retrieve data for multiple years. I don't want to gather too much data so I focused on the years that my arrivals by country dataset gave, which was 1995-2020. I ran into another issue, and that was that even though I was using `time.sleep()` in order to not send too many requests to the API in a short amount of time, I used the maximum requests I was allowed in a day. Luckily, I structured my document in a way that it kept appending the results each call, so I was able to pick up where I left off the following day. I think it would have been more practical to first clean up my prior datasets and determine exactly what countries I needed instead of requesting so much data, but I haven't yet determined exactly what I need for comparisons yet..

Spyder (Python 3.13)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\dream\Documents\School\CS332\Project\WeatherApi.py

MacJack_Module2_Code.py X WeatherApi.py X country_coords.csv X

```
58 country = row["Country"]
59 lat = row["Latitude"]
60 lon = row["Longitude"]
61
62 for year in range(1995, 2021):
63     if (country, year) in completed:
64         continue # Skip already collected entries
65
66     url = "https://archive-api.open-meteo.com/v1/archive"
67     params = {
68         "latitude": lat,
69         "longitude": lon,
70         "start_date": f"{year}-01-01",
71         "end_date": f"{year}-12-31",
72         "daily": [
73             "temperature_2m_mean",
74             "temperature_2m_max",
75             "temperature_2m_min",
76             "precipitation_sum",
77             "sunshine_duration",
78             "shortwave_radiation_sum"
79         ],
80         "timezone": "auto",
81         "temperature_unit": "fahrenheit",
82         "precipitation_unit": "mm"
83     }
84
85     try:
86         responses = openmeteo.weather_api(url, params=params)
87         response = responses[0]
88         daily = response.Daily()
89
90         # Extract and aggregate daily data into yearly summary
91         df = pd.DataFrame({
92             "date": pd.date_range(
93                 start=pd.to_datetime(daily.Time(), unit="s", utc=True),
94                 end=pd.to_datetime(daily.TimeEnd(), unit="s", utc=True),
95                 freq=pd.Timedelta(seconds=daily.Interval()),
96                 inclusive="left"
97             ),
98             "temperature_mean_f": daily.Variables(0).ValuesAsNumpy(),
99             "temperature_max_f": daily.Variables(1).ValuesAsNumpy(),
100             "temperature_min_f": daily.Variables(2).ValuesAsNumpy(),
101             "precipitation_mm": daily.Variables(3).ValuesAsNumpy(),
102             "sunshine_duration_s": daily.Variables(4).ValuesAsNumpy(),
103             "shortwave_radiation_mj": daily.Variables(5).ValuesAsNumpy()
104         })
105
106         summary = {
107             "country": country,
108             "year": year,
109             "avg_temp_mean_f": df["temperature_mean_f"].mean(),
110             "avg_temp_max_f": df["temperature_max_f"].mean(),
111             "avg_temp_min_f": df["temperature_min_f"].mean(),
112             "total_precip_mm": df["precipitation_mm"].sum(),
113             "avg_sunshine_hrs": df["sunshine_duration_s"].sum() / 3600,
114             "total_radiation_mj": df["shortwave_radiation_mj"].sum()
115         }
116
117         results.append(summary)
118         print(f"Retrieved data for {country}, {year}")
119
120     # Save every successful run immediately
121     pd.DataFrame(results).to_csv(output_file, mode="a", header=not os.path.exists(output_file), inc
122     results = [] # clear list after each write
123
124     time.sleep(5) # small delay between calls
125
126 except Exception as e:
127     print(f"Error for {country}, {year}: {e}")
128     print("Waiting 2 minutes before retry...")
129     time.sleep(120)
130     continue
```

Name	Type	Size	
cache_session	session.CachedSession	1	CachedSess
completed	set	0	{}
coords	DataFrame	[202, 3]	Column nam
country	str	22	British V
daily	VariablesWithTime.VariablesWithTime	1	Variables
df	DataFrame	[366, 7]	Column nam
final_df	DataFrame	[0, 0]	Column nam
lat	float	1	18.420695
lon	float	1	64.639968
openmeteo	Client.Client	1	Client obj
output_file	str	27	climate_d
params	dict	8	{'latitude
response	WeatherApiResponse.WeatherApiResponse	1	WeatherApi
responses	list	1	[WeatherAp
results	list	0	[]
retry_session	session.CachedSession	1	CachedSess
row	Series	[3]	Series obj
summary	dict	8	{'country'
url	str	45	https://ar
year	int	1	2012

Help Variable Explorer Deb

Console 3/A X

```
Retrieved data for Bhutan, 2002
Retrieved data for Bhutan, 2003
Retrieved data for Bhutan, 2004
Retrieved data for Bhutan, 2005
Retrieved data for Bhutan, 2006
Retrieved data for Bhutan, 2007
Retrieved data for Bhutan, 2008
Retrieved data for Bhutan, 2009
Retrieved data for Bhutan, 2010
Retrieved data for Bhutan, 2011
Retrieved data for Bhutan, 2012
Retrieved data for Bhutan, 2013
Retrieved data for Bhutan, 2014
Retrieved data for Bhutan, 2015
Retrieved data for Bhutan, 2016
Retrieved data for Bhutan, 2017
Retrieved data for Bhutan, 2018
Retrieved data for Bhutan, 2019
Retrieved data for Bhutan, 2020
Retrieved data for Bolivia, 1995
Retrieved data for Bolivia, 1996
```

IPython Console

Snippet of data set retrieved:

country	year	avg_temp_mean_f	avg_temp_max_f	avg_temp_min_f	total_precip_mm	avg_sunshine_hrs	total_radiation_mj
Afghanistan	1995	32.05074	41.002563	23.068647	306.3	3880.6956	7952.1006
Afghanistan	1996	33.12527	42.244843	24.002138	286.5	3930.1436	7991.54
Afghanistan	1997	31.392311	40.039936	22.51829	409.7	3834.756	7926.03
Afghanistan	1998	34.633327	43.208427	25.867275	302.7	3923.0476	7948.9995
Afghanistan	1999	34.642914	43.536865	25.495192	210.1	3895.5046	8004.33
Afghanistan	2000	34.48846	43.89804	24.76173	144.9	3984.6296	8151.84
Afghanistan	2001	35.72618	45.473225	25.966646	142.5	4015.5518	8163.63
Afghanistan	2002	34.781612	44.04481	25.36451	211.9	3939.1829	7978.2197
Afghanistan	2003	33.505318	42.64747	24.024866	225.20001	3940.4468	8007.93
Afghanistan	2004	35.26474	44.28632	25.71853	242.9	3938.3838	7931.0596
Afghanistan	2005	31.985498	40.379715	23.427168	356.5	3914.635	7927.9697
Afghanistan	2006	34.448788	43.003525	25.509003	363.1	3883.1196	7877.4
Afghanistan	2007	33.445484	41.772125	25.001057	337.9	3910.2947	7963.44
Afghanistan	2008	34.19299	42.796646	25.552794	259.2	3970.5618	8057.01
Afghanistan	2009	31.549133	39.585743	23.244453	416.10004	3879.369	7934.4805
Afghanistan	2010	35.263577	43.8764	26.731277	336.2	3922.6733	7795.6504
Afghanistan	2011	34.82846	43.670265	26.007084	290.3	3891.8306	7901.73
Afghanistan	2012	30.58432	38.872055	22.084106	436.30002	3890.7227	7867.0205
Afghanistan	2013	33.206604	41.594837	24.608015	317.5	3887.041	7967.11
Afghanistan	2014	32.56761	41.298454	23.891714	306.20004	3915.9314	7971.05
Afghanistan	2015	33.115288	41.621716	24.484728	447.7	3867.7466	7806.96
Afghanistan	2016	35.672386	44.231697	27.08989	326.30002	3937.6008	7925.51
Afghanistan	2017	32.070415	40.45021	23.47202	264.1	3867.4543	8179.9697
Afghanistan	2018	32.713425	41.44909	23.840649	206.90001	3894.334	8077.79
Afghanistan	2019	30.759945	38.515087	22.6418	334.19998	3765.3267	8091.18
Afghanistan	2020	30.48557	38.331505	21.99921	311.40002	3805.5518	8273.641
Albania	1995	56.510574	65.48846	47.232758	1041.9	3523.9133	5675.8496
Albania	1996	56.866665	65.52483	48.08008	1139.2001	3396.9395	5519.93
Albania	1997	57.15496	66.38944	47.63172	747.4	3558.201	5862.1396
Albania	1998	57.61448	66.660675	48.331005	973.6	3541.2886	5808.33
Albania	1999	58.16421	67.41618	48.694458	937.4	3542.5745	5697.7
Albania	2000	57.95542	68.0827	47.799995	708.19995	3690.7256	6038.2
Albania	2001	58.178013	67.3824	48.798267	896.30005	3608.0623	5790.84
Albania	2002	58.17682	67.38018	49.165913	961.1	3453.3906	5548.96
Albania	2003	58.73055	68.085884	48.967663	843.60004	3557.8384	5875.24
Albania	2004	57.80454	67.0536	48.397785	1117.7999	3490.2627	5662.66
Albania	2005	56.61234	65.69928	47.394264	1181.7	3510.1665	5671.34
Albania	2006	57.45334	67.00292	48.17912	846.6	3550.1453	5749.5303
Albania	2007	59.187084	68.725006	49.474865	759.1	3584.9473	5855.87
Albania	2008	58.86754	68.10434	49.42221	819.4	3600.883	5795.7397
Albania	2009	58.806625	67.76707	49.59421	1109.2	3490.5254	5683.76
Albania	2010	58.50443	67.023636	49.7032	1507.9	3380.905	5506.5
Albania	2011	58.311987	68.31355	48.260735	809	3685.205	6005.77

Data Cleaning

All Trails

Description: This dataset was manually gathered from the [Alltrails](#) website. It focuses on attributes of popular trails as well as overall ratings, trails counts, and review numbers by country. This data will be crucial in comparing tourism arrivals per country with top hiking destinations. This was not one that I was focusing on cleaning for submission on Module 5 & 6, but needed to finish cleaning before cleaning the following data sets as I wanted to narrow down what countries were actually needed. I will just run through this one without too many visualizations.

```
Country TrailCount HardPercent ... AvgElevGain_ft AvgRating ReviewCount
0 Australia 18,729 0.20 ... 1058.5 4.2 1253359
1 Austria 8,419 0.30 ... 1351.9 4.3 80138
2 Albania 59 0.70 ... 2562.1 4.3 5975
3 Algeria 4 0.50 ... 1140.5 4.4 23
4 Belize 36 0.25 ... 530.0 4.5 956

[5 rows x 8 columns]
```

Variables and Datatypes:

```
Data types of each column:
Country          object
TrailCount       object
HardPercent      float64
EasyPercent      float64
AvgLen_mi        float64
AvgElevGain_ft   float64
AvgRating        float64
ReviewCount      int64
dtype: object
```

Country - Object:

I confirmed that every country appears only once, so no duplicates. It is listed as an object which is correct.

TrailCount - object:

This is an incorrect data type and I am not sure why it is listed as an object but since they are all whole numbers, I changed it to an int64. I had to strip the “,” from it in order to change it to an int.

The remaining datatypes were the correct format. I confirmed that there were no missing entries.

Country Arrivals

Description: This dataframe contains the number of arrivals by year per country. It will assist in finding a correlation between the tourism numbers and popular hiking trails. We will also be able to see a trend of tourism highs and lows over a long stretch of time and determine if this is a possible influence of climate patterns.

Before Cleaning:

	A	B	C	D	E	F	G	H	I	J	K
1	Country Name	Country Code	Indicator Name	Indicator Code	1995	1996	1997	1998	1999	2000	2001
2	Aruba	ABW	International tou	ST.INT.ARVL	912000	957000	947000	906000	972000	1211000	1178000
3	Africa Eastern and Souther	AFE	International tou	ST.INT.ARVL	11583544.71	13088654.2	13456245.97	14403851.83	15309377.84	15353177.3	15854696.29
4	Afghanistan	AFG	International tou	ST.INT.ARVL							
5	Africa Western and Central	AFW	International tou	ST.INT.ARVL	2670705.934	3027134.733	3243144.347	3422652.337	3897975.031	4162850.003	4615887.342
6	Angola	AGO	International tou	ST.INT.ARVL	9000	21000	45000	52000	45000	51000	67000
7	Albania	ALB	International tou	ST.INT.ARVL	304000	287000	119000	184000	371000	317000	354000
8	Andorra	AND	International tou	ST.INT.ARVL					9422000	10991000	11351000
9	Arab World	ARB	International tou	ST.INT.ARVL							
10	United Arab Emirates	ARE	International tou	ST.INT.ARVL							
11	Argentina	ARG	International tou	ST.INT.ARVL	2289000	2614000	2764000	3012000	2898000	2909000	2620000
12	Armenia	ARM	International tou	ST.INT.ARVL	12000	13000	23000	32000	41000	45000	123000
13	American Samoa	ASM	International tou	ST.INT.ARVL	34000	35000	26000	36000	41000	44000	36000
14	Antigua and Barbuda	ATG	International tou	ST.INT.ARVL	447000	498000	526000	570000	560000	634000	624000
15	Australia	AUS	International tou	ST.INT.ARVL	3726000	4165000	4318000	4167000	4459000	4931000	4856000
16	Austria	AUT	International tou	ST.INT.ARVL	17173000	17090000	16647000	17352000	17467000	17982000	18180000
17	Azerbaijan	AZE	International tou	ST.INT.ARVL	93000	210000	306000	483000	602000	681000	767000
18	Burundi	BDI	International tou	ST.INT.ARVL	34000	27000	11000	15000	26000	29000	36000
19	Belgium	BEL	International tou	ST.INT.ARVL	5560000	5829000	6037000	6179000	6369000	6457000	6452000
20	Benin	BEN	International tou	ST.INT.ARVL	580000	516000	541000	575000	941000	1068000	1155000
21	Burkina Faso	BFA	International tou	ST.INT.ARVL	124000	131000	138000	160000	117000	126000	128000
22	Bangladesh	BGD	International tou	ST.INT.ARVL	156000	166000	182000	172000	173000	199000	207000
23	Bulgaria	BGR	International tou	ST.INT.ARVL	8005000	6811000	7543000	5240000	5056000	4922000	5104000
24	Bahrain	BHR	International tou	ST.INT.ARVL	2311000	1988000	2600000	2898000	3280000	3869000	4388000
25	Bahamas, The	BHS	International tou	ST.INT.ARVL	3239000	3416000	3446000	3348000	3648000	4204000	4183000
26	Bosnia and Herzegovina	BIH	International tou	ST.INT.ARVL			115000	148000	147000	171000	139000
27	Belarus	BLR	International tou	ST.INT.ARVL						2030000	1852000
28	Belize	BLZ	International tou	ST.INT.ARVL	342000	349000	305000	300000	340000	374000	385000
29	Bermuda	BMU	International tou	ST.INT.ARVL	557000	570000	562000	558000	549000	539000	458000
30	Bolivia	BOL	International tou	ST.INT.ARVL	284000	313000	355000	387000	342000	319000	316000
31	Brazil	BRA	International tou	ST.INT.ARVL	1991000	2666000	2850000	4818000	5107000	5313000	4773000
32	Barbados	BRB	International tou	ST.INT.ARVL	927000	957000	990000	1019000	947000	1078000	1035000
33	Brunei Darussalam	BRN	International tou	ST.INT.ARVL	498000	837000	643000	964000	967000	984000	840000
34	Bhutan	BTN	International tou	ST.INT.ARVL	4800.000191	5199.999809	5400.000095	6199.999809	7199.999809	7599.999905	6400.000095
35	Botswana	BWA	International tou	ST.INT.ARVL	636000	656000	765000	940000	1039000	1306000	1451000
36	Central African Republic	CAF	International tou	ST.INT.ARVL	26000	21000	17000	7500	10000	11199.99981	9899.999619
37	Canada	CAN	International tou	ST.INT.ARVL	41657000	43256000	45076000	48064000	49055000	48638000	47147000
38	Central Europe and the Bal	CEB	International tou	ST.INT.ARVL	188652644.9	199135866.3	203579244	200151256.8	200062962	206946379.3	182538824.4
39	Switzerland	CHE	International tou	ST.INT.ARVL	6946000	6730000	7039000	7185000	7154000	7821000	7455000

```
Country Name Country Code ... 2023 2024
0 Aruba ABW ... NaN NaN
1 Africa Eastern and Southern AFE ... NaN NaN
2 Afghanistan AFG ... NaN NaN
3 Africa Western and Central AFW ... NaN NaN
4 Angola AGO ... NaN NaN
.. ...
261 Kosovo XKX ... NaN NaN
262 Yemen, Rep. YEM ... NaN NaN
263 South Africa ZAF ... NaN NaN
264 Zambia ZMB ... NaN NaN
265 Zimbabwe ZWE ... NaN NaN

[266 rows x 34 columns]
```

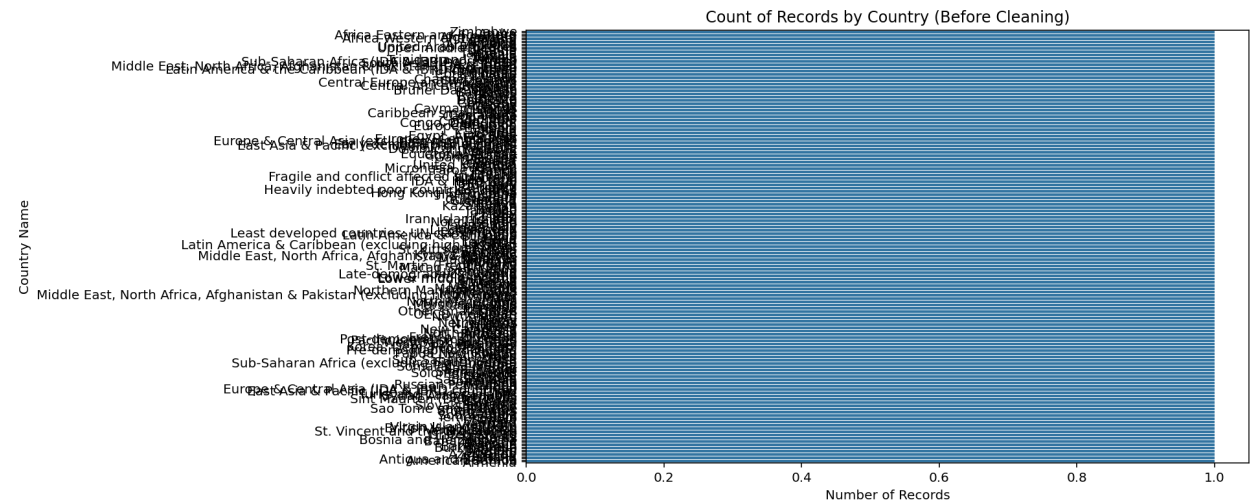
Variables and Datatypes:

```
Datatypes for Country Arrivals:

Country Name      object
Country Code      object
Indicator Name     object
Indicator Code     object
1960              float64
...
2021              float64
2022              float64
2023              float64
2024              float64
Unnamed: 69       float64
Length: 70, dtype: object
```

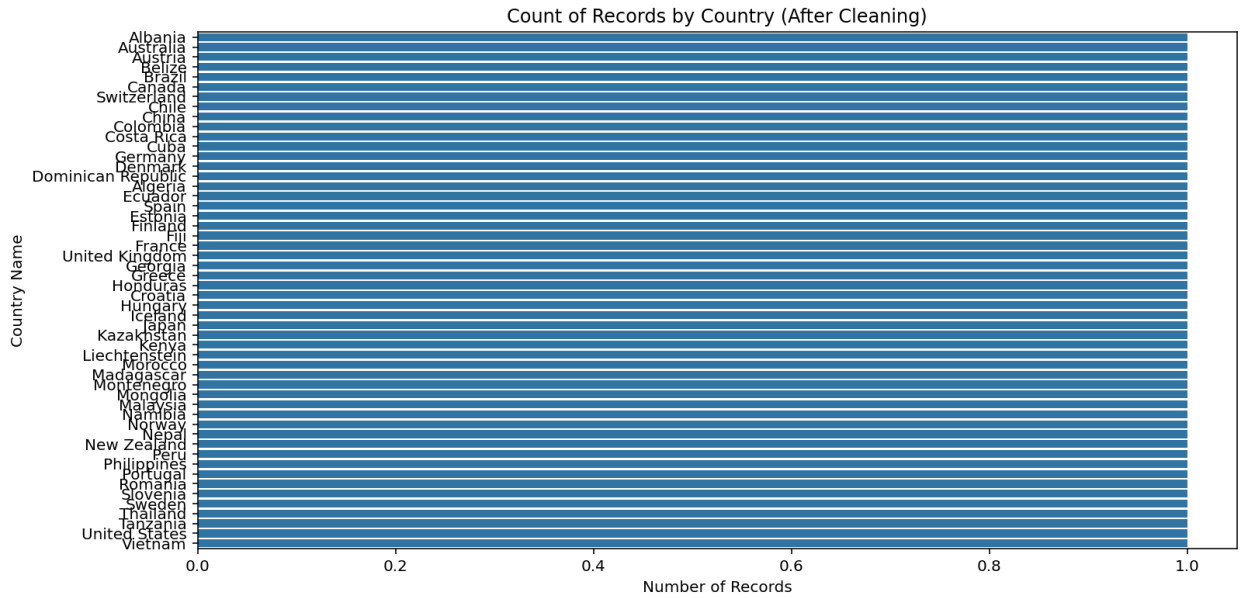
Country Name - object: This variable contains a number of countries with the correct data type of object.

Before Cleaning:



The visualization is very messy since there are a lot of countries. This does show me that there are no duplicates, as each country appears once. I believe that I will pull in my Alltrails data and only keep the countries that are listed in that dataframe, since I do not have any use for the other countries when performing my analysis. There are particularly some blends of countries that are labeled "IDA" or similar which are not specific enough for me to use, so those will certainly be removed. Every country has 1 record, so I do not need to check for duplicates but I will check for missing entries.

After Cleaning:



I removed the countries that were not listed in my all trails set. I checked for any missing entities with `isnull()`. This came back '0' so there was no need to correct it there. I needed to remove incorrect values such as "world", "high income" "IDA only" and so on, but after removing the entries that did not align with my all trails set, this took care of the problem. From the visualization, you can see that the countries were narrowed down quite a bit but we still have 52 countries. This also means only one country from my alltrails data was not found in the arrivals. One other piece of information this gives me is that the data must be in the correct format. They are capitalized and use their full names, otherwise the two sets wouldn't match up so easily.

Country Code - object: I will remove this column, as I do not have any use for it in analysis.

Indicator Name - object: This appears to be a column with repetitive data listing the name/description of the dataset. I will remove this since it is not needed.

Indicator Code - object: Again, I do not understand the purpose of this column so I will remove this. After removing 'Country Code', 'Indicator Name' and 'Indicator Code':

After removing unwanted values:

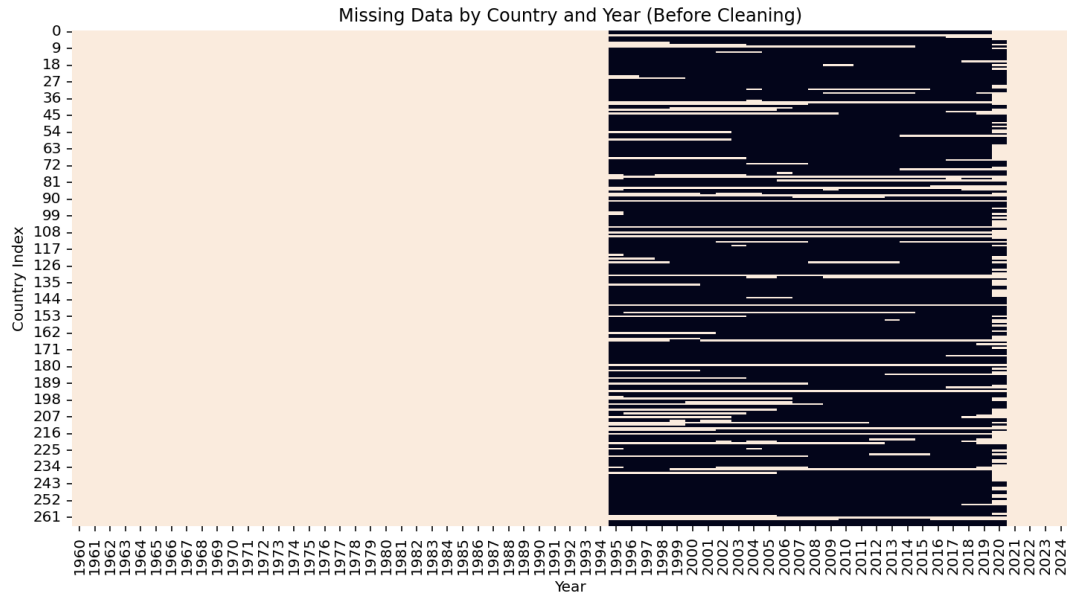
```
Datatypes for Country Arrivals

Country Name    object
1960            float64
1961            float64
1962            float64
1963            float64
...
2021            float64
2022            float64
2023            float64
2024            float64
Unnamed: 69     float64
Length: 67, dtype: object
```

Years Variables (1960-2020) - float64 : There is a column for every year from 1960 - 2020. I will check to see if there are missing entries within each year and look at outliers. This float value gives the number of arrivals per year for each country. I believe because some were likely averaged when collecting the data, that it is a float type. Because we have decimal numeric data, I will keep this type.

Before Cleaning:

```
Country Name    0
1960            266
1961            266
1962            266
1963            266
...
2021            266
2022            266
2023            266
2024            266
Unnamed: 69     266
Length: 67, dtype: int64
```

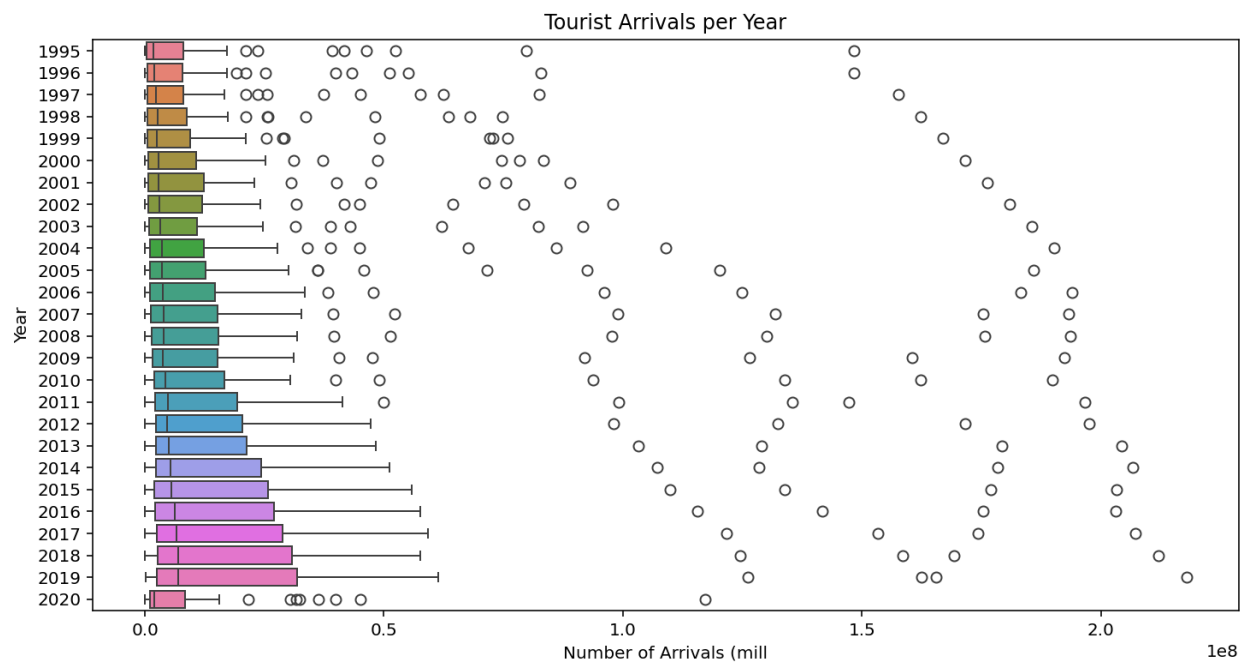


This tells me that there are some issues with empty data in the years columns. I will look at visualizations to better analyze this. I also produced a heatmap for every year which basically tells us the same thing about missing data. It shows us that there are a lot of outliers of years that have no data but are listed in the csv file. So everything but 1995-2020 will be dropped. It also shows that there are some countries and years that are missing quite a bit of data. I believe that after I keep only the countries that I need for analysis, it will narrow missing data down quite a bit.

After removing the countries that were not needed, as well as the years that were not in the 1995-2020 bracket, I found that my missing rows were significantly lessened. I thought that dropping rows could have severe consequences when I eventually want to see a trend or linear progression over a span of time so I looked into the best way to fill these missing values and found that an interpolate method of pandas could populate data while keeping that trend. Printing out missing values after doing this as well as another heatmap showed me that I had safely populated all missing data:

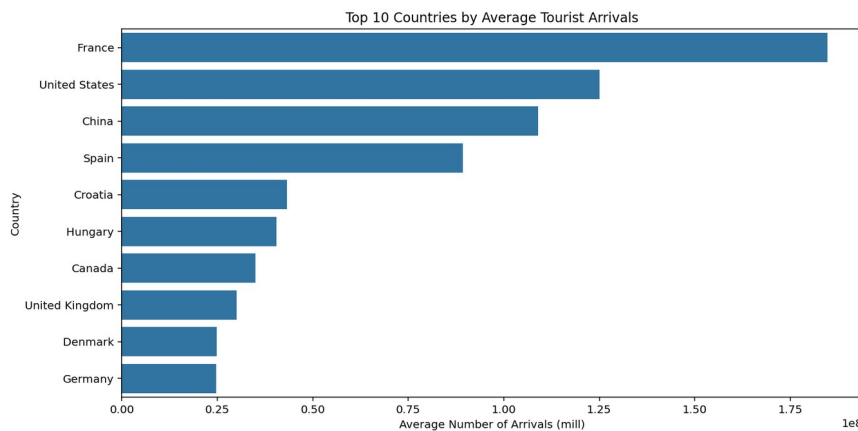


Now I want to check for outliers of every year or any values that need to be corrected. First, I printed out if there were any negative values across any of the year's columns. There were not, so I went to look at any extreme outliers. I have found one of the best ways to visualize outliers in this scenario is using a boxplot.



What this graph tells me is that there are a few extreme outliers every year. However, I believe that this is due to a few of the countries being extremely popular over others, so there will be some outliers. Also, we can see how the median line in each boxplot has a trend of shifting upward (to the right in this case), which tells us that there is a progression of arrivals per country over the years. As you can see, 2020 plunged down drastically. I do not believe that any of these outliers need to be removed, as it is expected that there will be some countries that will always have a larger amount of tourism (such as the United States). I think it will be important to consider these when analyzing our Alltrails data with this.

For one more thing to check, I decided to take the average arrivals over the years so we can see the top country arrivals. It might show us some of those outliers.



This confirms what I assumed earlier, that those outliers are some of the more heavily visited countries in the world. France not only is a large tourist destination, but there are many connecting flights and I am wondering if those are also considered in our dataset. The United States and China also being up there is no surprise to me.

After Cleaning Dataset:

	Country Name	1995	...	2019	2020
5	Albania	304000.0	...	6.406000e+06	2.658000e+06
13	Australia	3726000.0	...	9.466000e+06	1.828000e+06
14	Austria	17173000.0	...	3.188400e+07	1.509100e+07
26	Belize	342000.0	...	1.674000e+06	4.870000e+05
29	Brazil	1991000.0	...	6.353000e+06	6.353000e+06
35	Canada	41657000.0	...	3.243000e+07	3.243000e+07
37	Switzerland	6946000.0	...	1.181800e+07	1.181800e+07
39	Chile	3713000.0	...	5.431000e+06	5.431000e+06
40	China	46387000.0	...	1.625380e+08	3.040200e+07
45	Colombia	1433000.0	...	4.531000e+06	1.396000e+06
48	Costa Rica	924000.0	...	3.366000e+06	1.146500e+06
50	Cuba	745000.0	...	4.276000e+06	1.086000e+06
55	Germany	14847000.0	...	3.956300e+07	1.244900e+07
58	Denmark	21119000.0	...	3.309300e+07	1.559500e+07
59	Dominican Republic	1806000.0	...	7.550000e+06	2.748000e+06
60	Algeria	520000.0	...	2.371000e+06	5.910000e+05
66	Ecuador	440000.0	...	2.108000e+06	2.108000e+06
70	Spain	52460000.0	...	1.261700e+08	3.641000e+07
71	Estonia	2111000.0	...	6.103000e+06	1.695000e+06
75	Finland	1779000.0	...	3.290000e+06	8.960000e+05
76	Fiji	476000.0	...	9.690000e+05	1.680000e+05
77	France	148263000.0	...	2.178770e+08	1.171090e+08
81	United Kingdom	23537000.0	...	4.085700e+07	1.110100e+07
82	Georgia	85000.0	...	7.726000e+06	1.513000e+06
89	Greece	10712000.0	...	3.400500e+07	7.406000e+06
97	Honduras	271000.0	...	2.315000e+06	6.690000e+05
99	Croatia	16100000.0	...	6.002100e+07	2.160800e+07
101	Hungary	39240000.0	...	6.139700e+07	3.164100e+07
114	Iceland	211000.0	...	2.202000e+06	4.880000e+05
119	Japan	3345000.0	...	3.188100e+07	4.115800e+06
120	Kazakhstan	202000.0	...	8.515000e+06	2.035000e+06
121	Kenya	974000.0	...	2.049000e+06	2.049000e+06
137	Liechtenstein	59000.0	...	9.810000e+04	5.840000e+04
148	Morocco	2752000.0	...	1.310900e+07	2.802000e+06
151	Madagascar	115000.0	...	4.860000e+05	8.710000e+04
162	Montenegro	136000.0	...	2.510000e+06	3.510000e+05
163	Mongolia	108000.0	...	6.370000e+05	6.690000e+04
169	Malaysia	7469000.0	...	2.610100e+07	4.333000e+06
171	Namibia	421000.0	...	1.651000e+06	1.871000e+05
177	Norway	2880000.0	...	5.879000e+06	1.397000e+06
178	Nepal	363000.0	...	1.197000e+06	2.300000e+05
180	New Zealand	1409000.0	...	3.888000e+06	9.960000e+05
186	Peru	1650000.0	...	5.275000e+06	1.119000e+06
187	Philippines	1760000.0	...	8.261000e+06	1.483000e+06
194	Portugal	9511000.0	...	1.728300e+07	4.208000e+06
201	Romania	5445000.0	...	1.281500e+07	5.023000e+06
222	Slovenia	732000.0	...	4.702000e+06	1.216000e+06
223	Sweden	2310000.0	...	7.616000e+06	1.957000e+06
233	Thailand	6952000.0	...	3.991600e+07	3.991600e+07
246	Tanzania	295000.0	...	1.527000e+06	1.527000e+06
251	United States	79732000.0	...	1.654780e+08	4.503700e+07
257	Vietnam	1351000.0	...	1.800900e+07	3.837000e+06

[52 rows x 27 columns]

Normalizing Arrivals Data

Before Normalizing Data:

```
Before Normalizing:
Country Name      1995  ...      2019      2020
5      Albania    304000.0  ...    6406000.0  2658000.0
13     Australia  3726000.0  ...    9466000.0  1828000.0
14     Austria    17173000.0  ...   31884000.0  15091000.0
26     Belize     342000.0  ...    1674000.0   487000.0
29     Brazil     1991000.0  ...    6353000.0   6353000.0
35     Canada     41657000.0  ...   32430000.0  32430000.0
37     Switzerland  6946000.0  ...   11818000.0  11818000.0
39     Chile      3713000.0  ...    5431000.0   5431000.0
40     China      46387000.0  ...  162538000.0  30402000.0
45     Colombia   1433000.0  ...    4531000.0   1396000.0
48     Costa Rica  924000.0  ...    3366000.0   1146500.0
50     Cuba       745000.0  ...    4276000.0   1086000.0
55     Germany    14847000.0  ...   39563000.0  12449000.0
58     Denmark    21119000.0  ...   33093000.0  15595000.0
59  Dominican Republic  1806000.0  ...   7550000.0   2748000.0
60     Algeria     520000.0  ...    2371000.0   591000.0
66     Ecuador    440000.0  ...    2108000.0   2108000.0
70     Spain      52460000.0  ...  126170000.0  36410000.0
71     Estonia    2111000.0  ...    6103000.0   1695000.0
75     Finland    1779000.0  ...   3290000.0    896000.0

[20 rows x 27 columns]
```

After Normalizing Data:

```
After normalization:
Country Name      1995      1996  ...      2018      2019      2020
5      Albania    0.001653  0.001559  ...  0.027567  0.028965  0.022209
13     Australia  0.024743  0.027725  ...  0.043229  0.043016  0.015118
14     Austria    0.115476  0.114934  ...  0.145016  0.145955  0.128428
26     Belize     0.001910  0.001977  ...  0.007605  0.007236  0.003662
29     Brazil     0.013036  0.017611  ...  0.030841  0.028721  0.053777
35     Canada     0.280681  0.291484  ...  0.147177  0.148462  0.276561
37     Switzerland  0.046470  0.045032  ...  0.054880  0.053816  0.100466
39     Chile      0.024655  0.024675  ...  0.030757  0.024488  0.045900
40     China      0.312596  0.344599  ...  0.748047  0.745894  0.259235
45     Colombia   0.009271  0.004689  ...  0.020347  0.020355  0.011428
48     Costa Rica  0.005837  0.005965  ...  0.015231  0.015006  0.009296
50     Cuba       0.004629  0.006396  ...  0.021833  0.019184  0.008779
55     Germany    0.099781  0.102215  ...  0.183074  0.181215  0.105857
58     Denmark    0.142101  0.142119  ...  0.144945  0.151506  0.132734
59  Dominican Republic  0.011788  0.013366  ...  0.035230  0.034218  0.022978
60     Algeria     0.003111  0.003704  ...  0.012136  0.010437  0.004550
66     Ecuador    0.002571  0.002955  ...  0.011560  0.009229  0.017510
70     Spain      0.353573  0.371244  ...  0.586896  0.578899  0.310563
71     Estonia    0.013846  0.016113  ...  0.028067  0.027573  0.013982
75     Finland    0.011606  0.011255  ...  0.014811  0.014657  0.007156

[20 rows x 27 columns]
```

Prepping for Kmeans

Before Prepping Data:

```
After normalization:
Country Name      1995      1996      ...      2018      2019      2020
5      Albania  0.001653  0.001559  ...  0.027567  0.028965  0.022209
13      Australia 0.024743  0.027725  ...  0.043229  0.043016  0.015118
14      Austria  0.115476  0.114934  ...  0.145016  0.145955  0.128428
26      Belize   0.001910  0.001977  ...  0.007605  0.007236  0.003662
29      Brazil   0.013036  0.017611  ...  0.030841  0.028721  0.053777
35      Canada   0.280681  0.291484  ...  0.147177  0.148462  0.276561
37      Switzerland 0.046470  0.045032  ...  0.054880  0.053816  0.100466
39      Chile    0.024655  0.024675  ...  0.030757  0.024488  0.045900
40      China    0.312596  0.344599  ...  0.748047  0.745894  0.259235
45      Colombia 0.009271  0.004689  ...  0.020347  0.020355  0.011428
48      Costa Rica 0.005837  0.005965  ...  0.015231  0.015006  0.009296
50      Cuba     0.004629  0.006396  ...  0.021833  0.019184  0.008779
55      Germany  0.099781  0.102215  ...  0.183074  0.181215  0.105857
58      Denmark  0.142101  0.142119  ...  0.144945  0.151506  0.132734
59      Dominican Republic 0.011788  0.013366  ...  0.035230  0.034218  0.022978
60      Algeria  0.003111  0.003704  ...  0.012136  0.010437  0.004550
66      Ecuador  0.002571  0.002955  ...  0.011560  0.009229  0.017510
70      Spain    0.353573  0.371244  ...  0.586896  0.578899  0.310563
71      Estonia  0.013846  0.016113  ...  0.028067  0.027573  0.013982
75      Finland  0.011606  0.011255  ...  0.014811  0.014657  0.007156

[20 rows x 27 columns]
```

After Prepping Data:

```
After Prepping for Kmeans:|
      1995      1996      1997      ...      2018      2019      2020
5      0.001653  0.001559  0.000394  ...  0.027567  0.028965  0.022209
13      0.024743  0.027725  0.027055  ...  0.043229  0.043016  0.015118
14      0.115476  0.114934  0.105337  ...  0.145016  0.145955  0.128428
26      0.001910  0.001977  0.001575  ...  0.007605  0.007236  0.003662
29      0.013036  0.017611  0.017734  ...  0.030841  0.028721  0.053777
35      0.280681  0.291484  0.285846  ...  0.147177  0.148462  0.276561
37      0.046470  0.045032  0.044332  ...  0.054880  0.053816  0.100466
39      0.024655  0.024675  0.023214  ...  0.030757  0.024488  0.045900
40      0.312596  0.344599  0.365290  ...  0.748047  0.745894  0.259235
45      0.009271  0.004689  0.004451  ...  0.020347  0.020355  0.011428
48      0.005837  0.005965  0.006070  ...  0.015231  0.015006  0.009296
50      0.004629  0.006396  0.007067  ...  0.021833  0.019184  0.008779
55      0.099781  0.102215  0.100194  ...  0.183074  0.181215  0.105857
58      0.142101  0.142119  0.133732  ...  0.144945  0.151506  0.132734
59      0.011788  0.013366  0.015397  ...  0.035230  0.034218  0.022978
60      0.003111  0.003704  0.003670  ...  0.012136  0.010437  0.004550
66      0.002571  0.002955  0.002997  ...  0.011560  0.009229  0.017510
70      0.353573  0.371244  0.395939  ...  0.586896  0.578899  0.310563
71      0.013846  0.016113  0.016261  ...  0.028067  0.027573  0.013982
75      0.011606  0.011255  0.011270  ...  0.014811  0.014657  0.007156

[20 rows x 26 columns]
```

Climate_data.csv

Description: This data was gathered with the use of an API, 'Open-Meteo's Historical Weather'. A separate CSV that listed countries and their center coordinates in order to achieve data by country through the API.

Before Cleaning:

country	year	avg_temp_mear	avg_temp_max	avg_temp_min_f	total_precip_mm	avg_sunshine_h	total_radiation_r
Afghanistan	1995	32.05074	41.002563	23.068647	306.3	3880.6956	7952.1006
Afghanistan	1996	33.12527	42.244843	24.002138	286.5	3930.1436	7991.54
Afghanistan	1997	31.392311	40.039936	22.51829	409.7	3834.756	7926.03
Afghanistan	1998	34.633327	43.208427	25.867275	302.7	3923.0476	7948.9995
Afghanistan	1999	34.642914	43.536865	25.495192	210.1	3895.5046	8004.33
Afghanistan	2000	34.48846	43.89804	24.76173	144.9	3984.6296	8151.84
Afghanistan	2001	35.72618	45.473225	25.966646	142.5	4015.5518	8163.63
Afghanistan	2002	34.781612	44.04481	25.36451	211.9	3939.1829	7978.2197
Afghanistan	2003	33.505318	42.64747	24.024866	225.20001	3940.4468	8007.93
Afghanistan	2004	35.26474	44.28632	25.71853	242.9	3938.3838	7931.0596
Afghanistan	2005	31.985498	40.379715	23.427168	356.5	3914.635	7927.9697
Afghanistan	2006	34.448788	43.003525	25.509003	363.1	3883.1196	7877.4
Afghanistan	2007	33.445484	41.772125	25.001057	337.9	3910.2947	7963.44
Afghanistan	2008	34.19299	42.796646	25.552794	259.2	3970.5618	8057.01
Afghanistan	2009	31.549133	39.585743	23.244453	416.10004	3879.369	7934.4805
Afghanistan	2010	35.263577	43.8764	26.731277	336.2	3922.6733	7795.6504
Afghanistan	2011	34.82846	43.670265	26.007084	290.3	3891.8306	7901.73
Afghanistan	2012	30.58432	38.872055	22.084106	436.30002	3890.7227	7867.0205
Afghanistan	2013	33.206604	41.594837	24.608015	317.5	3887.041	7967.11
Afghanistan	2014	32.56761	41.298454	23.891714	306.20004	3915.9314	7971.05
Afghanistan	2015	33.115288	41.621716	24.484728	447.7	3867.7466	7806.96
Afghanistan	2016	35.672386	44.231697	27.08989	326.30002	3937.6008	7925.51
Afghanistan	2017	32.070415	40.45021	23.47202	264.1	3867.4543	8179.9697
Afghanistan	2018	32.713425	41.44909	23.840649	206.90001	3894.334	8077.79
Afghanistan	2019	30.759945	38.515087	22.6418	334.19998	3765.3267	8091.18
Afghanistan	2020	30.48557	38.331505	21.99921	311.40002	3805.5518	8273.641
Albania	1995	56.510574	65.48846	47.232758	1041.9	3523.9133	5675.8496
Albania	1996	56.866665	65.52483	48.08008	1139.2001	3396.9395	5519.93
Albania	1997	57.15496	66.38944	47.63172	747.4	3558.201	5862.1396

\Before Cleaning Climate Data:

```
country year ... avg_sunshine_hrs total_radiation_mj
0 Afghanistan 1995 ... 3880.6956 7952.1006
1 Afghanistan 1996 ... 3930.1436 7991.5400
2 Afghanistan 1997 ... 3834.7560 7926.0300
3 Afghanistan 1998 ... 3923.0476 7948.9995
4 Afghanistan 1999 ... 3895.5046 8004.3300
...
4952 Zimbabwe 2016 ... 4093.8410 8910.3000
4953 Zimbabwe 2017 ... 4008.1736 8694.2810
4954 Zimbabwe 2018 ... 3995.2808 8568.3810
4955 Zimbabwe 2019 ... 3988.4434 8504.2210
4956 Zimbabwe 2020 ... 4007.4683 8560.2000
```

[4957 rows x 8 columns]

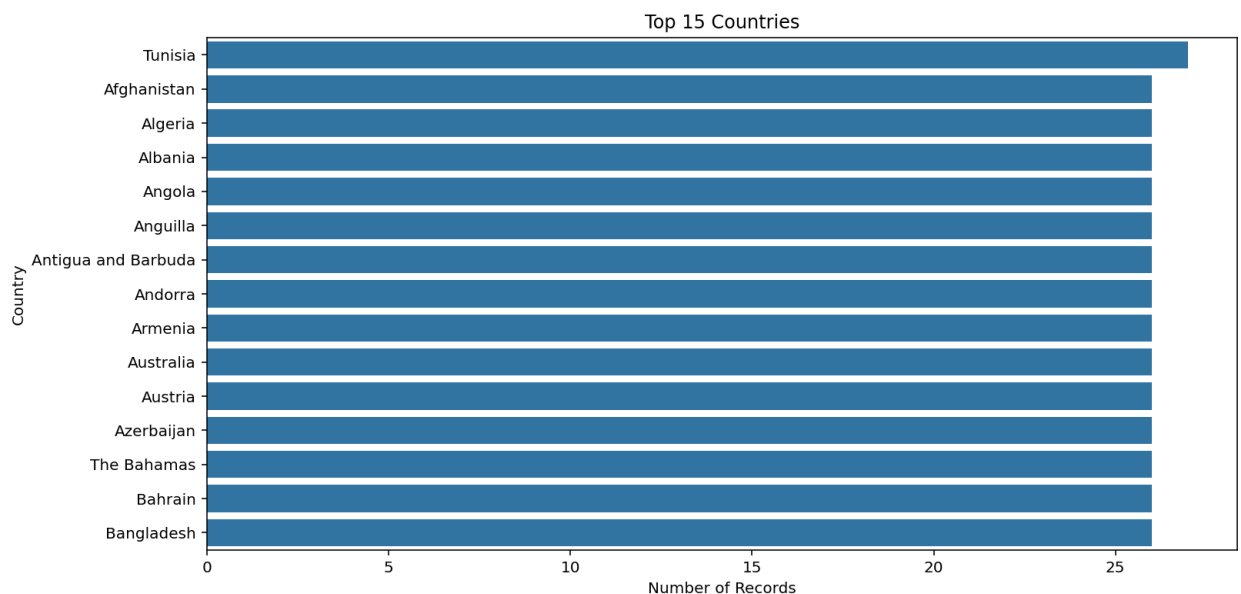
Variables and Datatypes:

```
country      object
year         int64
avg_temp_mean_f  float64
avg_temp_max_f  float64
avg_temp_min_f  float64
total_precip_mm  float64
avg_sunshine_hrs float64
total_radiation_mj float64
dtype: object
```

Country - object: This variable is an object that holds multiple entries of the same country, so there will be duplicates.

Before Cleaning:

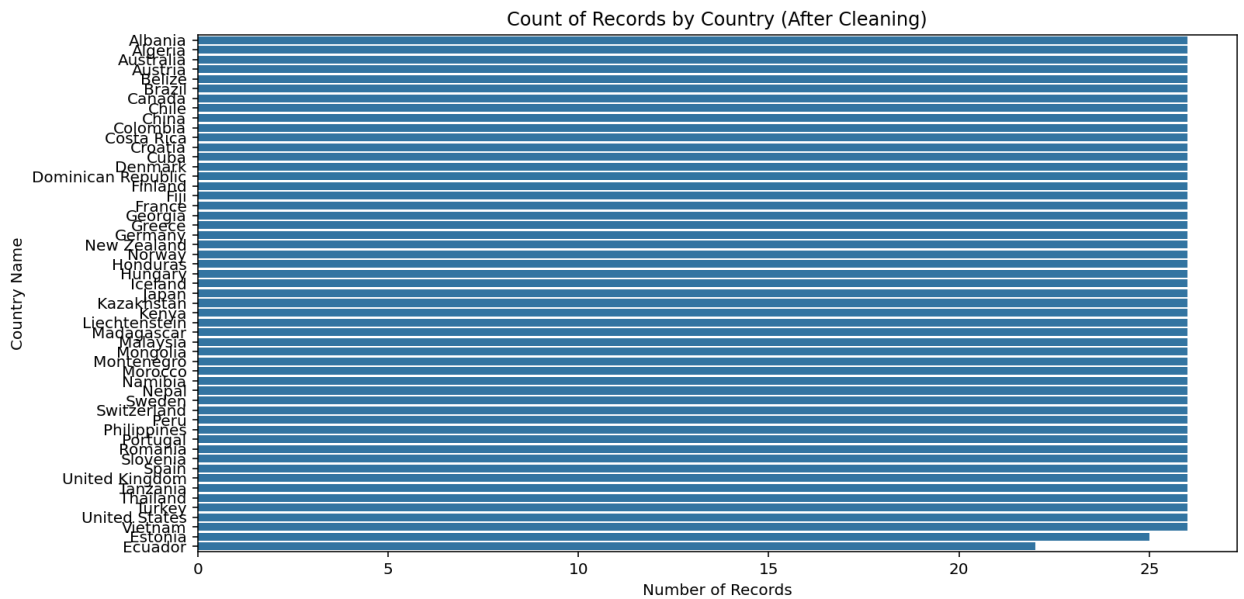
My one concern was that because I pulled this data with an API and needed to do multiple runs due to the length of it, I need to make sure that I have an equal amount of country duplicates. Each country should be listed from 1995-2020. I also want to check for any missing (null entries) and filter out the countries that are not on my all trails set.



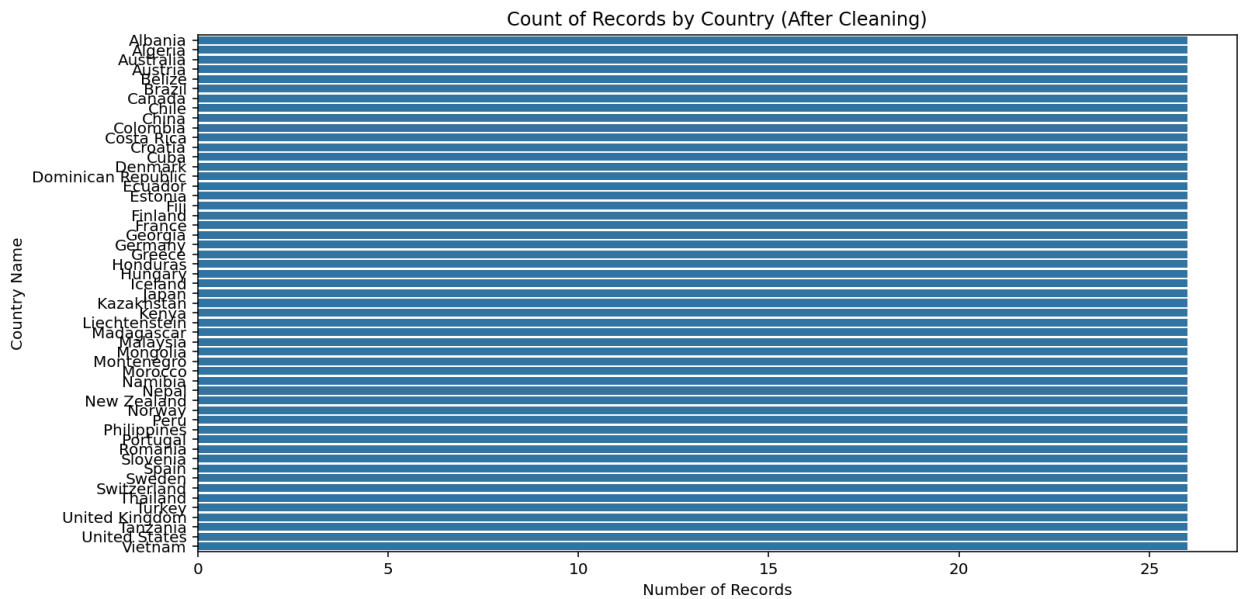
I went with a histogram and frequency count for each country name. The before visualization tells me that Tunisia has a higher number of records than the rest of the countries so it is likely I accidentally have one repetitive row. Just like with my previous dataset, once I eliminate all of the countries that I will not be using in my analysis, it will be much easier to manage.

I began by checking for any missing entries in this column, which came back zero. By the visualization I can see that my country names are formatted the same as my other data set (capitalized). Then I filtered out the countries that are not in my all trails set again.

After Cleaning:



I found that I was still missing my entries, likely some years that I missed when pulling from the API. So I had to go back and fill in those missing years.

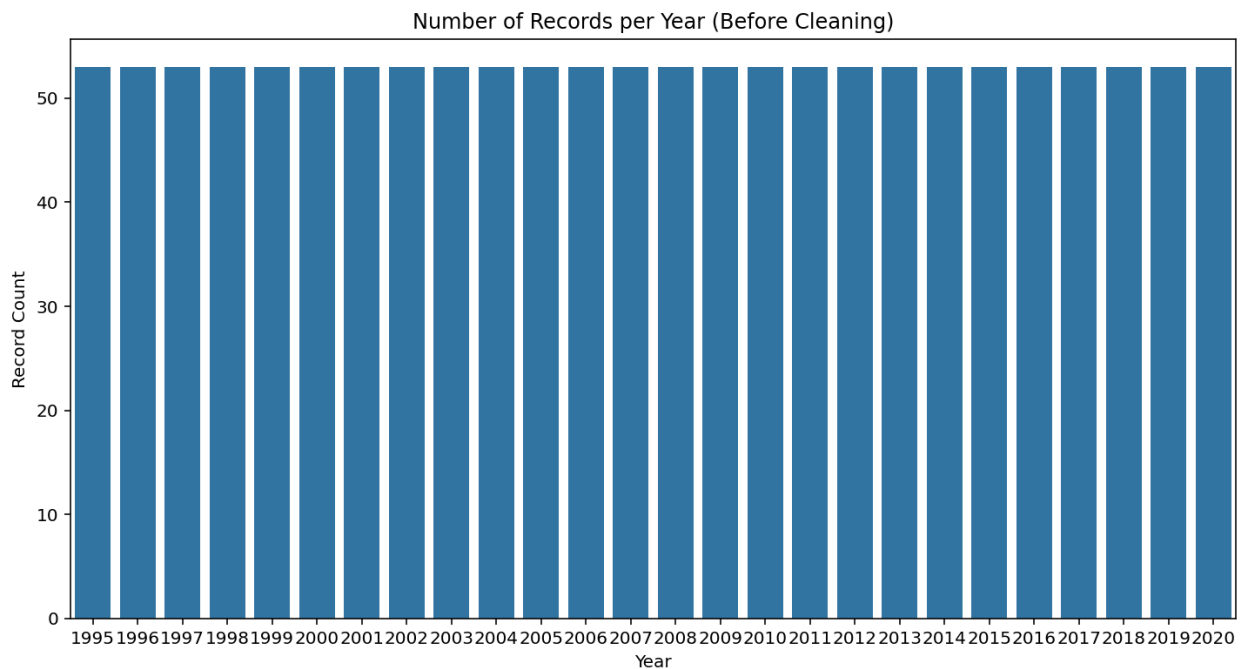


So now I have no missing records, correct formatting, and only the countries I need.

year-int 64: This variable represents the mean annual temperature of every country. It is formatted as a float, which is the correct data type.

Before Cleaning:

This column has the correct data type. I had only pulled records from 1995-2020 through the api but I will double check that there are no outliers or duplicates. I actually took care of this when cleaning my country data, as I deciphered that there were some missing years by the number of country records that were reported.



After Cleaning:

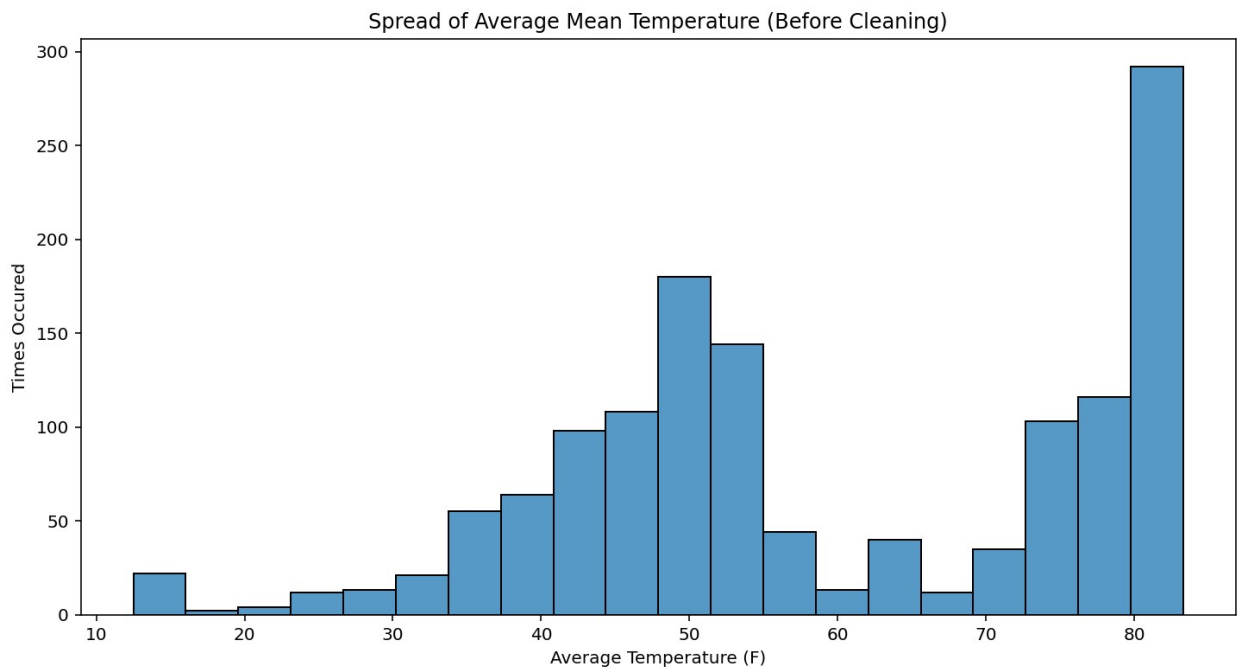
I do not believe there is anything left to clean in this variable, especially after I filled in those missing data when cleaning my countries. The visualization shows that we have the same amount of records (53) from 1995-2020, which is to be expected because my Alltrails dataset has 53 countries. So we have no missing records, no outliers, no wrong years, etc.

Avg_temp_mean_f - float: This variable represents the mean annual temperature per country. It is correctly stored as a float.

Before Cleaning:

The most important things I want to focus on when cleaning is if there are any missing entries, or if there are any outliers. I googled what would be an expected temperature range and found that the average global temperature is 59 degrees fahrenheit, while there can be an expected range of about 10-90 degrees fahrenheit. If I do find any odd data, I will likely use the interpolate method again since we are looking at a linear progression over time. Because we are also dealing with multiple averages for duplicate

years and country entries, I wanted to see the count of temperatures spread out so I can get an idea of the frequency of hotter or cooler degree averages.



We are in the range of the 10-90 degree mark, which shows that there are no major outliers (negatives or very large numbers). Our second largest range is right around the average global temp of 59, which again supports data accuracy. Although there is a concentration of very high values (80-90 range), these are likely hotter regions and not data errors. I also used `describe()` in order to get statistics on the mean, percentiles, etc. This further supports my visualization:

```
count    1378.000000
mean      59.254375
std       17.717526
min       12.496732
25%       46.588361
50%       54.000747
75%       78.711915
max       83.312675
Name: avg_temp_mean_f, dtype: float64
```

No missing, negative, or any invalid values were found. Since the data matches up with global climate averages, there is no need to interpolate or clean further.

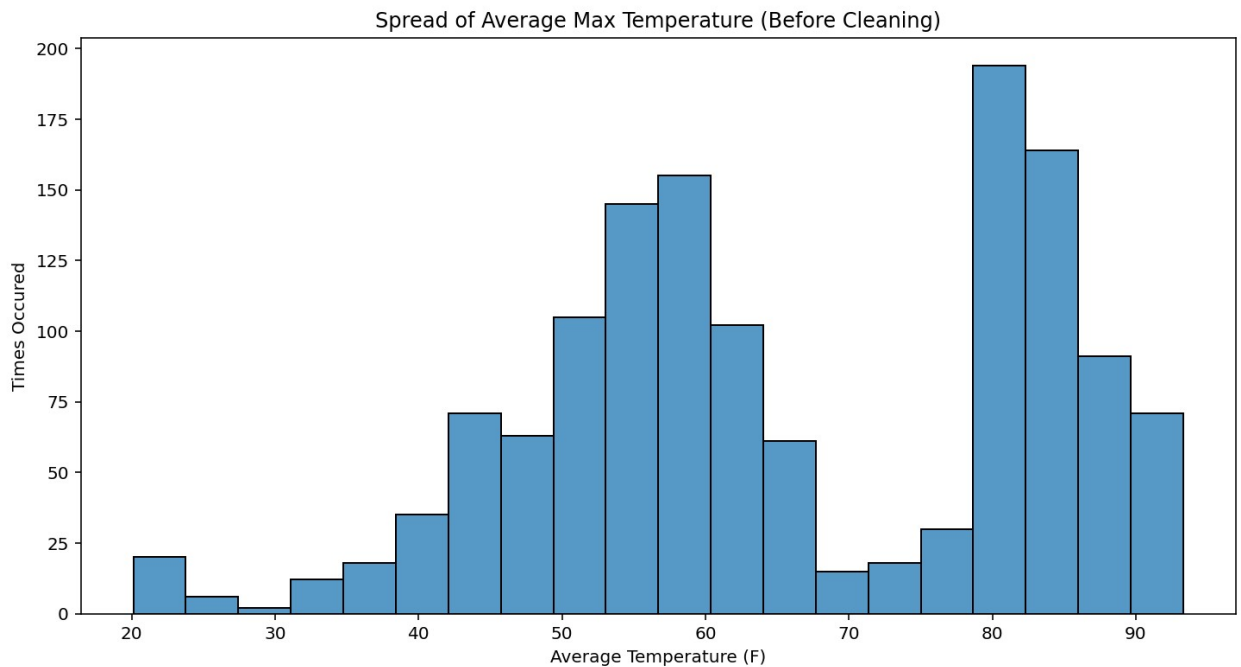
After Cleaning:

This variable did not require any further cleaning. I kept the data type as a float64, and my visualization of distribution matched up to realistic climate reports. I also did not find any outlier to remove or input any missing data.

Avg_temp_max_f - float64: this variable represents the average of the daily maximum temperatures recorded for each country and year. The data type is correctly stored as a float.

Before Cleaning:

Similar to what I did with the avg_tem_mean variable, I expect that while cleaning I will check for any missing values. I will look at possible outliers that are in an expected temp range. I looked up what the coldest and hottest temperatures recorded were and found that it was 134 degrees F and -128. I think anything in our data that is close to these numbers should require further analysis, and anything outside of this range is impossible.



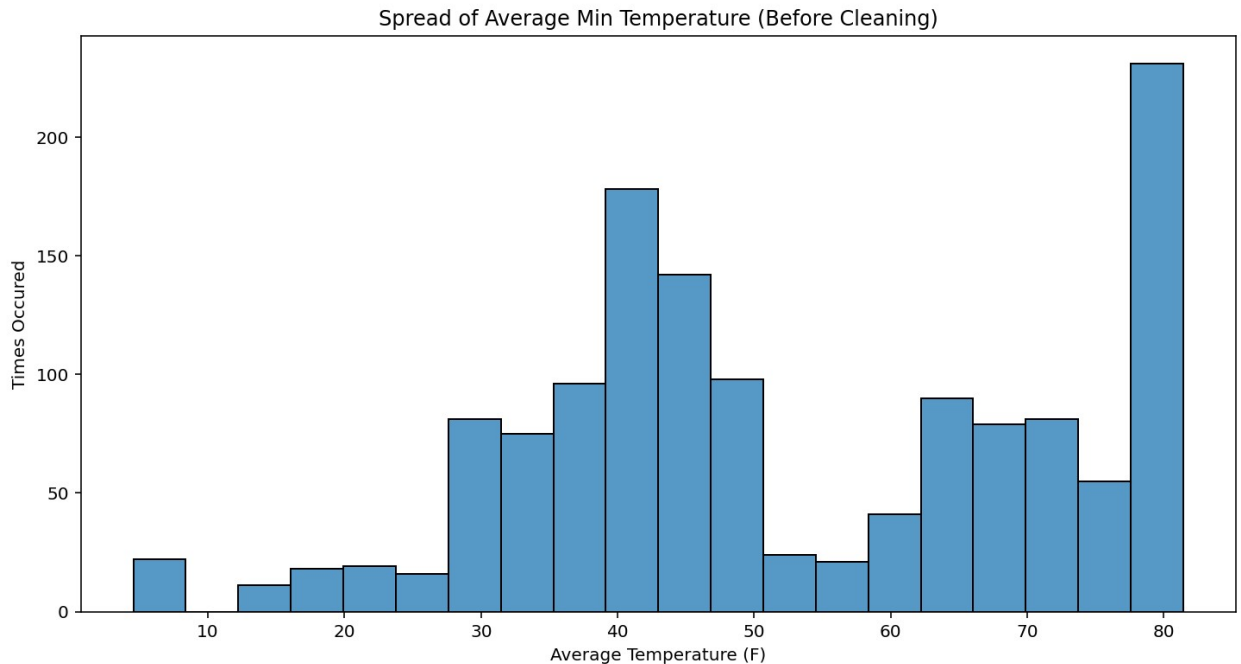
After Cleaning:

All of the values for this variable fall within a realistic temperature range (20 F - 100 F), as you can see in the visualization. We have a correct float time and there are also no missing or invalid entries. This was confirmed by the visualization and writing code to check for missing entries. Because the data represents a realistic maximum temps, no further cleaning is required.

Avg_temp_min_f - float64: This variable represents the average daily minimum temperatures for each country and year. The data is stored as a float type, which is correct.

Before Cleaning:

As with the previous variables, I want to check for any missing entries, outliers, and incorrect formatting. A realistic temperature range, as listed in the previous variable, is -128 F to 134 F. I believe that because we are dealing with a minimum temp, any values above around 100 would be near impossible and anything below -128 would also be impossible. If this occurs in our data, this will likely need to be corrected.



At first, I was a little worried not to see any negative numbers on the visualization. However, once looking back at all of my data, I remembered that the Open-Meteo API is reporting the average minimum daily temperature, so will likely not show any cold or hot spikes. This reports that the majority of the annual min temps fall into a realistic range of 30-70 degrees. It shows that there is no missing data (confirmed with additional code) or impossible outliers.

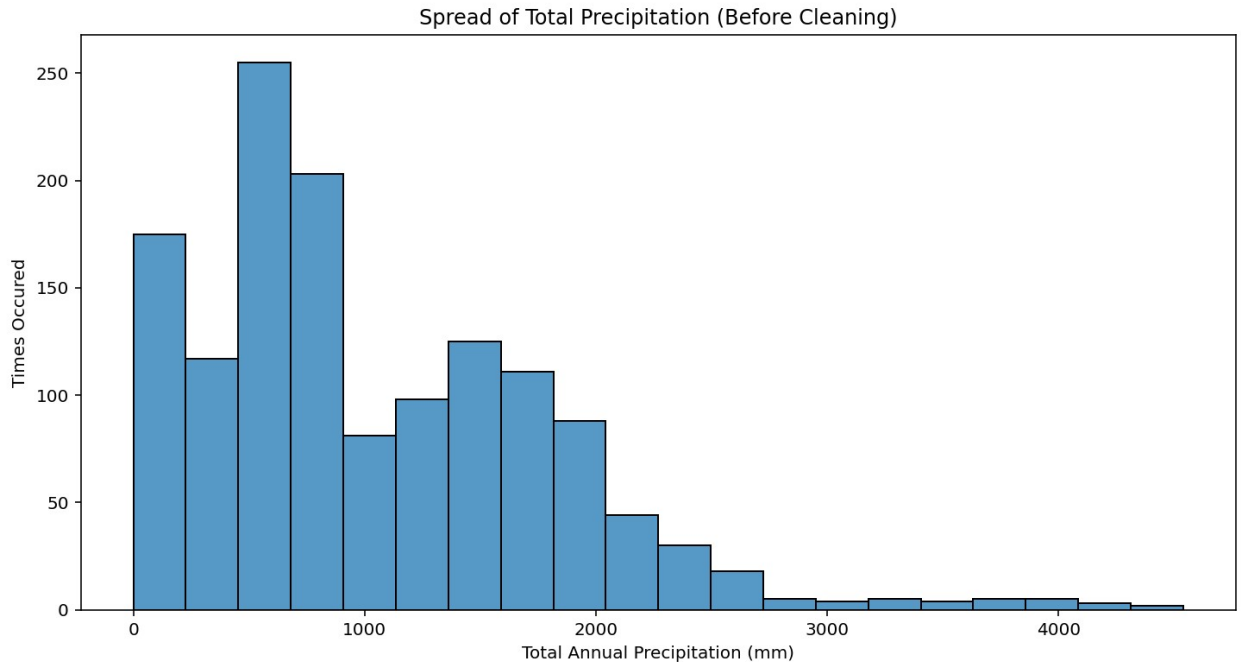
After Cleaning:

No corrections were needed. All of the data fell in a realistic range for the annual daily minimums, data type was in the correct form (float), and there were no missing or odd data.

Total_precip_mm - float64: this variable represents the total annual precipitation (in millimeters) for each country. It is correctly stored as a float type.

Before Cleaning:

I will need to check for any missing values, negative entries, or extremely large totals. Upon further research, I found that the global precipitation range annually falls somewhere between 0 - 10000 mm. We are dealing with more desert regions on the lower precipitation end and much more tropical wet areas on the higher end. We can use this range to identify any outliers.



I checked for any missing entries and confirmed there were none. The visualization shows that there are no negative number reports and that all data falls into a realistic range of 0-5000 mm. There are a few higher totals, but those likely correspond to more tropical regions with a lot of rainfall, such as Vietnam or Thailand. I was curious about this and decided to group the mean by country and sort them, so I could see what were the countries with the most precipitation.

country	
Philippines	3270.392300
Nepal	2812.126962
Malaysia	2251.823108
Ecuador	2030.461542
Switzerland	2012.492300
Montenegro	1974.380781
Belize	1957.165388
Peru	1907.353842
New Zealand	1800.803842
Honduras	1782.646173

I forgot that the Philippines were in my data set, so this makes perfect sense that they would be in the top. I was wrong about Vietnam and Thailand though.

Similarly for driest countries:

Namibia	1.903846
Madagascar	14.280769
Algeria	23.388462
Morocco	31.003847
Brazil	45.061539
United States	53.553846
Kazakhstan	220.469233
Mongolia	275.711544
Turkey	399.246151
Spain	454.792309

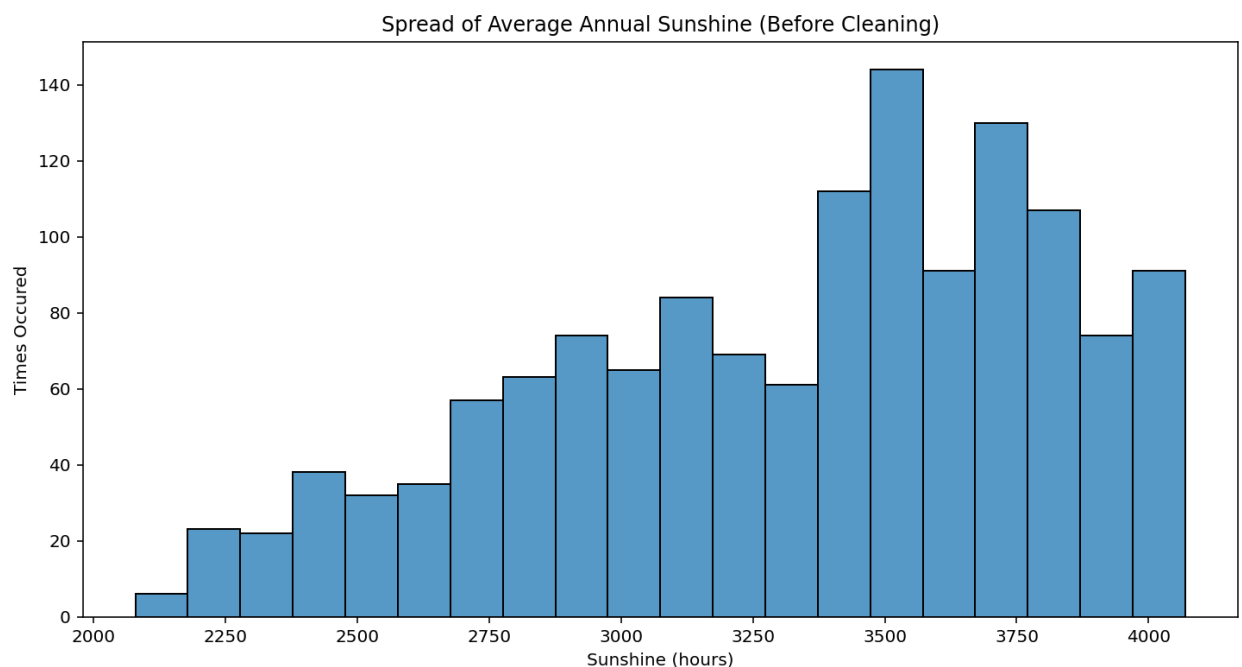
After Cleaning:

I checked for missing entries as well negative outliers or extreme large outliers. I confirmed that we had a correct data type of a float, and that all of the totals fell into a realistic range according to global mins/maximums. No further cleaning was required.

Avg_sunshine_hrs - float64: This variable represents the total annual sunshine duration (in hours). It is correctly stored as a float data type.

Before Cleaning:

Again, I will check for any missing entries. I will also look at any outliers so I need to determine what a good range for this might be. I suppose the maximum possible daylight hours could be on the higher end, while 0 would be on the lower end. The maximum possible daylight hours per year possible would be 8760 hours (by doing some simple math). Being close to this is extremely unlikely, but exceeding it is impossible. So I will need to check for anything outside of 0 - 8760 and if I do find something on the higher range, investigate the location it is occurring.



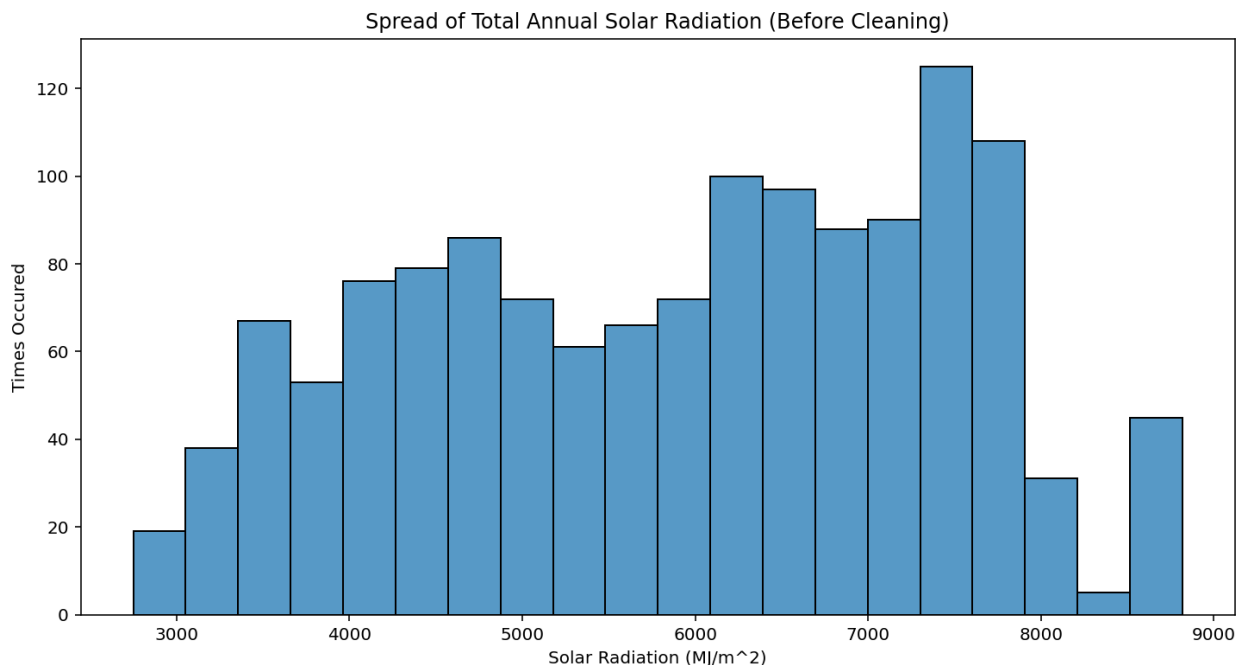
My histogram visualization shows that most countries fall in that 2000-4000 range. There are no major outliers of negative numbers or numbers close to (or exceeding) the maximum possible of 8760. I also checked for any missing entries. As you can see by the plot, all entries are numeric.

After Cleaning:

I verified that there were no missing data and that data was in the correct format. I also found that there were no anomalies of over 8760, or any negative numbers. Because all records fell in my expected range and there were no missing values, no further cleaning was necessary.

Total_radiation_mj - float64: This variable represents the total annual solar radiation, or solar energy, in megajoules. This is per square meter. It has the correct data type of a float.

As with my previous variables, I will check to see if there are any missing entries, outliers, or odd formats. I do not know what a typical outlier should look like for solar radiation, but do know we can't have any negative numbers. After doing research, I found that there are some desert locations where it is typical to have a high annual solar radiation of 8000 mj per square meter. So I think that a good range to check would be 0 - 10000, though the higher end might warrant some further investigating.



The histogram shows us the majority of the values are within that 6000-8000 mark. There were little values below 3000 and above 8000, which makes them a realistic representation of global radiation coverage. There were zero negatives, and though there were a few outliers of a typical high of 8000, not so high that it would be impossible. I checked to see what some of the higher radiation reports were located and if it matched up what we know about that region:

Top Countries with High Radiation:

country	
Madagascar	8616.992058
Namibia	8616.292638
Algeria	7986.251542
Brazil	7795.590788
Morocco	7750.129531
Fiji	7736.274277
Kenya	7722.614577
Chile	7693.549958
Colombia	7429.848438
Ecuador	7394.945692

This closely lines up with the statistics of our driest climates and appears to be realistic.

After Cleaning:

I verified that all values were stored as floats. I identified that there were no impossible values that needed to be removed. I also checked to see if there were any missing errors. None of these factors required me to clean the data any further.

After Cleaning Data Set:

After Cleaning Climate Data:

	country	year	...	avg_sunshine_hrs	total_radiation_mj
26	Albania	1995	...	3523.9133	5675.8496
27	Albania	1996	...	3396.9395	5519.9300
28	Albania	1997	...	3558.2010	5862.1396
29	Albania	1998	...	3541.2886	5808.3300
30	Albania	1999	...	3542.5745	5697.7000
...	...	...	...	...	...
4879	Vietnam	2016	...	3487.4448	6546.8496
4880	Vietnam	2017	...	3385.9534	6465.1997
4881	Vietnam	2018	...	3501.0125	6783.4795
4882	Vietnam	2019	...	3676.6438	7053.0000
4883	Vietnam	2020	...	3620.9119	7038.4897

[1378 rows x 8 columns]

Unsupervised Learning with KMeans Clustering

AllTrails Data

Prepping for KMeans

In the previous section, the data set was cleaned. In order to prepare the datasets for KMeans clustering, I needed the variables to be numerically comparable across all of my features, so applied the min-max normalization.

Before Normalizing Data Set:

```
Original AllTrails Data (First 5 Rows):
```

	Country	TrailCount	HardPercent	...	AvgElevGain_ft	AvgRating	ReviewCount
2	Albania	59	0.70	...	2562.1	4.3	5975
3	Algeria	4	0.50	...	1140.5	4.4	23
0	Australia	18729	0.20	...	1058.5	4.2	1253359
1	Austria	8419	0.30	...	1351.9	4.3	80138
4	Belize	36	0.25	...	530.0	4.5	956

[5 rows x 8 columns]

After Normalizing Data Set:

```
After Normalization (First 5 Rows):
```

	Country	TrailCount	HardPercent	...	AvgElevGain_ft	AvgRating	ReviewCount
2	Albania	0.000514	0.70	...	0.074156	0.375	0.000171
3	Algeria	0.000027	0.50	...	0.024894	0.500	0.000000
0	Australia	0.165865	0.20	...	0.022053	0.250	0.035930
1	Austria	0.074554	0.30	...	0.032220	0.375	0.002297
4	Belize	0.000310	0.25	...	0.003739	0.625	0.000027

[5 rows x 8 columns]

Next, I needed to remove any columns that were not quantitative.

After removal:

```
AllTrails Data Prepared for KMeans (First 5 Rows):
```

	TrailCount	HardPercent	EasyPercent	...	AvgElevGain_ft	AvgRating
ReviewCount						
2	0.000514	0.70	0.125	...	0.074156	0.375
0.000171						
3	0.000027	0.50	0.625	...	0.024894	0.500
0.000000						
0	0.165865	0.20	0.250	...	0.022053	0.250
0.035930						
1	0.074554	0.30	0.125	...	0.032220	0.375
0.002297						
4	0.000310	0.25	0.375	...	0.003739	0.625
0.000027						

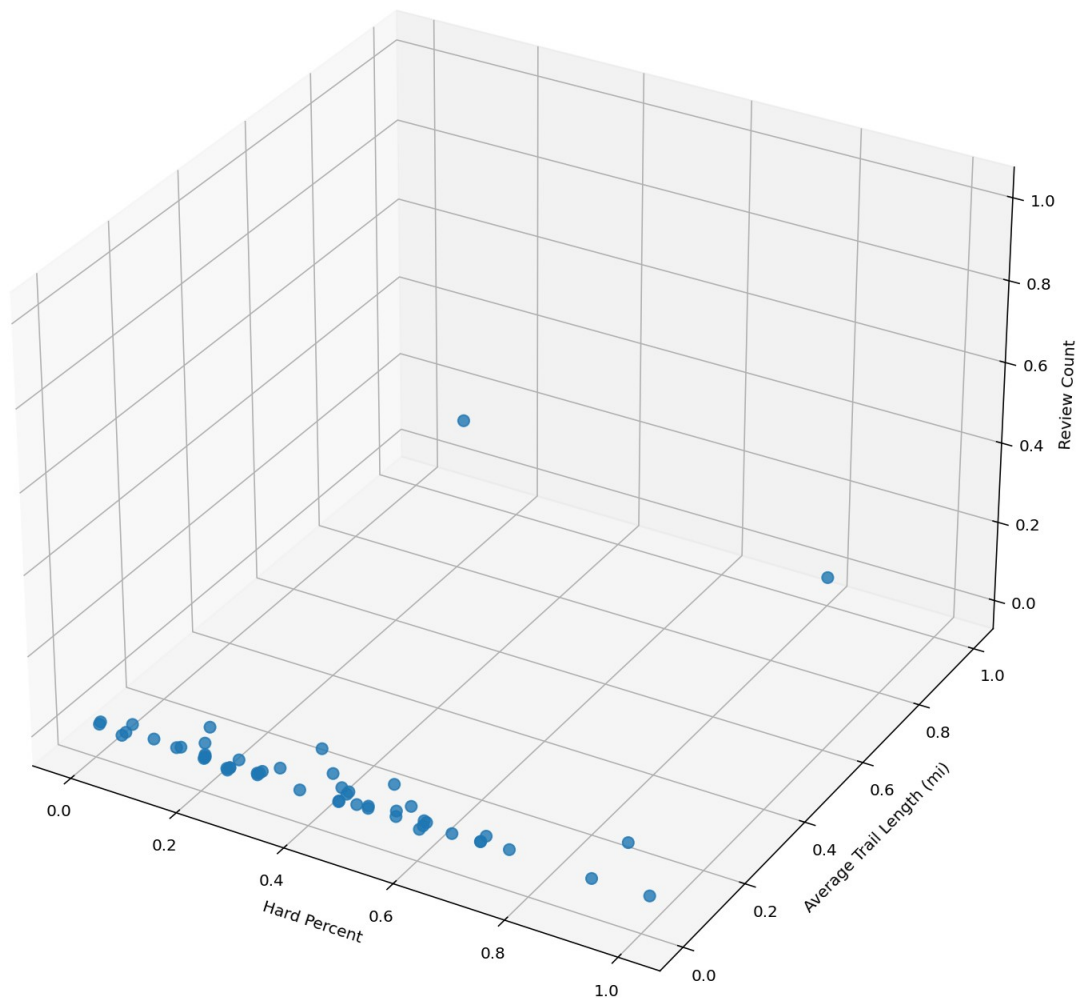
```
[5 rows x 7 columns]
```

Now that non numerical columns, such as “country” are removed, KMeans can analyze my quantitative variables. I prepped all three of my datasets for KMeans, but ultimately decided to focus on the Alltrails dataset for this step in the project.

Visualizing Data

Now that my data is completely prepared for a KMeans analysis, I needed to choose three variables to perform a visual analysis on. I still have several columns available to me: TrailCount, HardPercent, EasyPercent, AvgLen_mi, AvgElevGain_ft, AvgRating, and ReviewCount. I decided to look at the trail difficulty vs length vs popularity, so I took the columns: HardPercent, AvgLen_mi, and ReviewCount.

Trail Difficulty, Length, and Popularity (Before Clustering)



Each point in the scatterplot represents one country. The three chosen variables represent the difficulty of hiking trails, their average length, and overall popularity by review count. We have the data normalized with the min-max method, so the differences in scale did not distort the relationship between these variables. Most countries appear to group near the origin (the lower values of all three in the left hand side). This shows that many nations have smaller trail systems with lower difficulty and fewer total reviews.

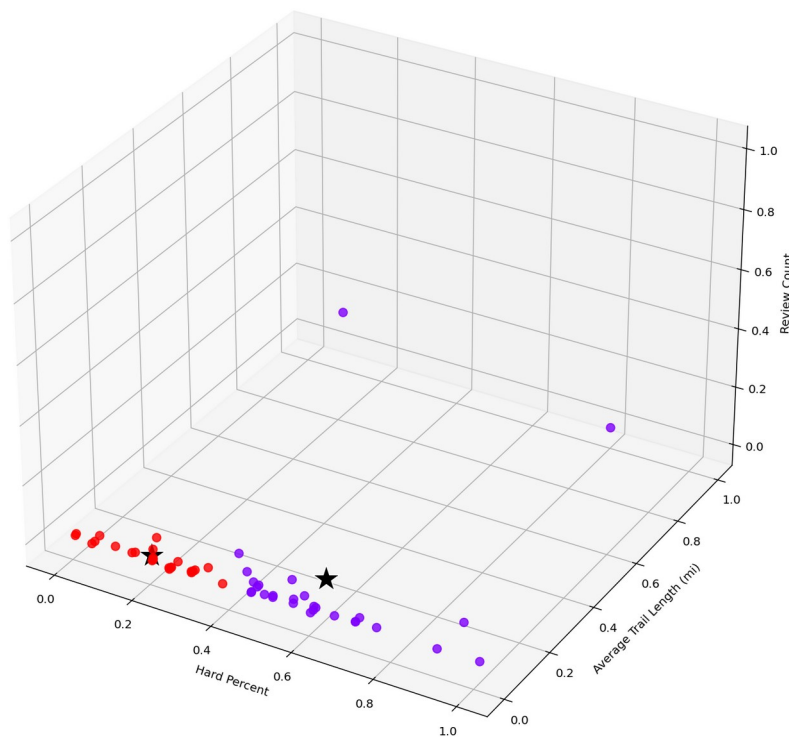
There are a few points that extend far along the ReviewCount and AvgLen_mi axes. These are likely countries such as The United States, Canada, and Australia. They indicate much greater trail popularity along our scale. There is an uneven distribution here but I do believe that this shows us there are some divisions that exist among countries that have different hiking characteristics.

Apply KMeans

K = 2

```
Cluster assignments (labels):  
[0 0 1 1 1 0 0 0 0 0 1 1 1 1 0 0 1 0 1 1 0 1 1 0 1 1 0 0 0 0 0 1 0 0 1 1 0  
 0 0 0 0 1 0 1 0 1 0 1 1 0 1 1 1 0 1]  
  
Cluster centroids:  
[[0.60246032 0.0685349 0.04114524]  
 [0.1968     0.01505641 0.0073159 ]]  
  
Predicted cluster [0.5, 8, 50000]: [0]
```

Trail Difficulty, Length, and Popularity (Before Clustering)



Here, the data separates into two very broad groups. From the visualization, it appears that one group contains the countries with fewer trails, lower difficulty percentages, and less total reviews overall. The other cluster represents countries that have a higher amount of difficult trails and longer trail lengths, with more views. The centroids appear to show that these groups separate into regions that are less developed or more established in terms of hiking activity. I also predicted a new data

vector by inserting a new theoretical data point: This new data represented a country with a 50 hard percent, 8 miles, and 50000 reviews. The prediction put this example in cluster 0, which is the cluster that has a more active hiking presence.

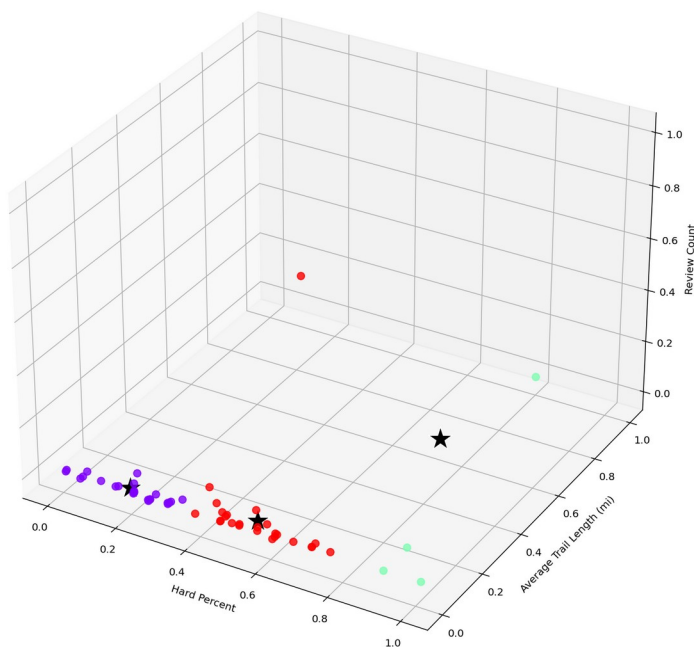
K = 3

```
Cluster assignments (labels):
[2 2 0 0 0 2 2 2 2 0 0 0 0 2 2 0 2 0 0 2 0 0 2 0 0 2 2 2 2 0 1 2 0 0 1
 2 2 1 2 0 2 0 2 0 0 1 2 0 0 2 0]

Cluster centroids:
[[1.89166667e-01 1.53027424e-02 7.58489979e-03]
 [9.00000000e-01 2.96250786e-01 2.61906159e-04]
 [5.45955556e-01 2.97247401e-02 4.60751565e-02]]

Predicted cluster [0.5, 8, 50000]: [2]
```

Trail Difficulty, Length, and Popularity (Before Clustering)



Increasing k to 3 resulted in three distinct groups spanned across the plot. The cluster to the left near the origin represents countries with relatively easy trails with shorter lengths and fewer reviews. These countries likely have smaller hiking cultures. The middle cluster contains countries with moderate difficulty percentages but also shows an extreme outlier with longer trail and review count. The cluster on the far right contains only a few data points, representing countries with much more difficulty percentages and possible more challenging terrain. Overall, it appears that k = 3 shows

more of a scale from casual hikes to advanced. The far right cluster also shows us that countries with near exclusively difficult trails are very rare.

I also see that our new data point was predicted to be in cluster 2, which shows that it resembles more of a moderate hiking destination.

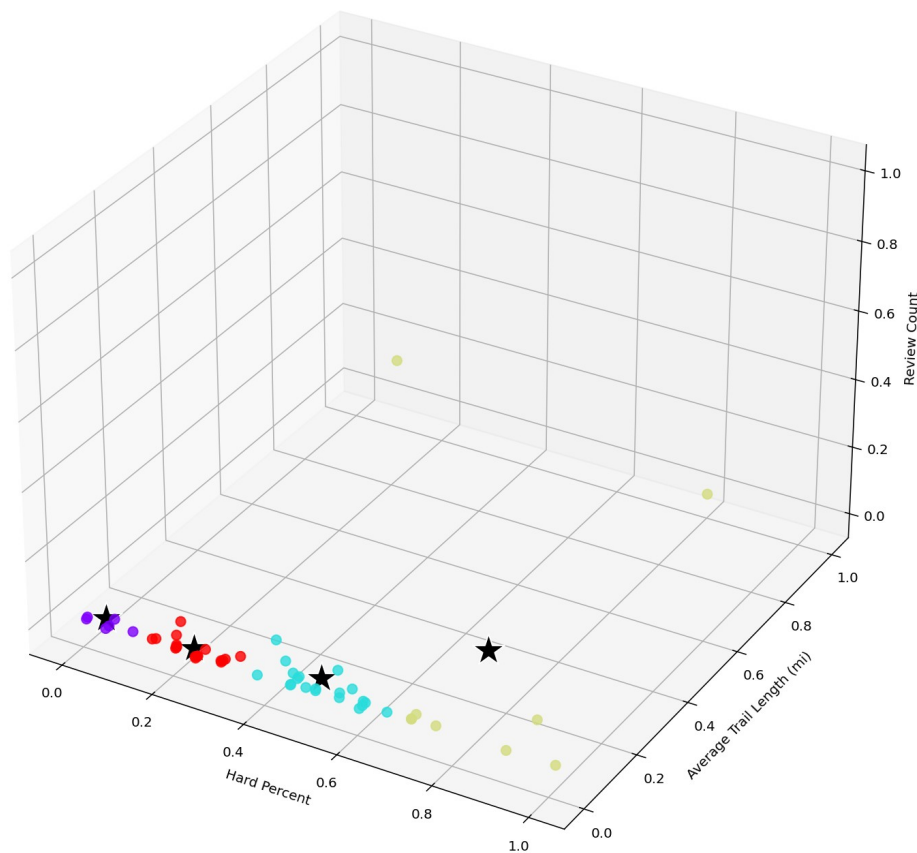
K = 4

```
Cluster assignments (labels):
[2 1 3 3 3 1 1 1 2 1 0 3 3 0 1 1 0 1 3 3 2 3 3 1 0 3 1 1 1 1 1 3 2 2 3 0 2
 1 1 2 1 3 1 3 1 0 3 2 1 3 3 2 3]

Cluster centroids:
[[0.04166667 0.01667905 0.00169298]
 [0.50594444 0.03031075 0.00755294]
 [0.79222222 0.14687851 0.11131865]
 [0.23833333 0.01484397 0.00954887]]

Predicted cluster [0.5, 8, 50000]: [2]
```

Trail Difficulty, Length, and Popularity (Before Clustering)



Increasing K further to 4 separated the dataset into even more compact tighter groups. Two of the clusters seem to overlap my earlier “moderate trail” region. This indicates that the middle of this distribution might have some very subtle variations such as some countries lean toward slightly easier trails with shorter distances, while others are lean more toward moderate or slightly longer hikes. The very left cluster, close to the origin, still represents areas with very low hard percent values, shorter trails, and low review counts. The very far right now groups the two very extreme outliers with much longer trail lengths and review counts, most of them being more difficult terrain. I think these boundaries start to segment similar data points and that $k = 3$ provides a more meaningful clustering.

Conclusion

Through KMeans clustering, I found that countries naturally group into a few main trail patterns. Most countries fall into a broad, or moderate range of trail difficulty and length, while a smaller group shows much harder trails and more reviews (user activity). Increasing the number of clusters beyond three did not appear to give more useful insights which confirms that $k = 3$ better represents trends of trail difficulty and activity.

One other thing I have taken away from these visualizations, is that there does not appear to be that much of a correlation between countries and their difficulty level vs their trail length. Meaning, just because a trail was labeled as difficult does not mean it necessarily has to have a very long trail length (besides a couple of outliers). Many countries show a mix of shorter and moderate distance trails within higher difficult levels.

Overall, we can take away that global hiking regions can be divided into three categories: beginner friendly/accessible, balanced mix of easy/difficult, and advanced/challenging hikes. This sort of data could help the travel and tourism industry have a better understanding of where to focus certain marketing efforts, such as promoting trails that are easier in regions that are typically adventure-esque or improving accessibility in regions where trails are typically very difficult.

Supervised Learning w/ Decision Trees

Before Image:

AllTrails_Labeled ☆ 📁 ☁									
File Edit View Insert Format Data Tools Extensions Help									
🔍 Menus ↶ ↷ 🖨 🗑 100% ▾ \$ % ⬇ ⬆ 123 Default... ▾ - 10 + B I ↺ A ↻ 🏠 📄 📊									
M21 ▾	fx								
	A	B	C	D	E	F	G	H	I
1	Country	TrailCount	HardPercent	EasyPercent	AvgLen_mi	AvgElevGain_ft	AvgRating	ReviewCount	TrailLevel
2	Albania	59	0.7	0.05	6.93	2562.1	4.3	5975	Low Trails
3	Algeria	4	0.5	0.25	19.7	1140.5	4.4	23	Low Trails
4	Australia	18729	0.2	0.1	5.21	1058.5	4.2	1253359	High Trails
5	Austria	8419	0.3	0.05	5.66	1351.9	4.3	80138	High Trails
6	Belize	36	0.25	0.15	2.81	530	4.5	956	Low Trails
7	Brazil	9527	0.45	0.1	4.86	1748.5	4.5	27756	High Trails
8	Canada	29814	0.42	0.1	4.925	1710.05	4.2	4085482	High Trails
9	Chile	2280	0.45	0.15	10.02	2449.2	4.4	39530	Moderate Trails
10	China	548	0.75	0.1	7.48	29280	4.3	7924	Moderate Trails
11	Colombia	2047	0.55	0.05	6.05	1681.9	4.5	13714	Moderate Trails
12	Costa Rica	1589	0.05	0.15	3.01	624	4.4	30115	Moderate Trails
13	Croatia	4192	0.2	0.2	4.99	917.7	4.3	46101	Moderate Trails
14	Cuba	21	0.25	0.3	8.8	1169.5	4.6	152	Low Trails
15	Denmark	9079	0.05	0.1	8.42	787.2	4.1	191757	High Trails
16	Dominican Repu	81	0.4333333333	0.1	11.04	3062.9	4.5	1903	Low Trails
17	Ecuador	227	0.6	0.1	7.71	5801.6	4.4	9330	Low Trails
18	Estonia	262	0	0.35	4.84	422.1	4.3	3309	Low Trails
19	Fiji	9	0.5555555556	0.1111111111	12.54444444	11272.4	4.6	255	Low Trails
20	Finland	5561	0.2	0.1	4.48	701	4.2	18772	High Trails
21	France	39816	0.34	0	5.3025	1507.65	4.2	1154426	High Trails
22	Georgia	208	0.7	0	10.02	3312.2	4.4	8691	Low Trails
23	Germany	37885	0.15	0.25	4.06	1010.8	4.1	282913	High Trails
24	Greece	3787	0.3	0.05	4.52	1598.7	4.1	83454	Moderate Trails
25	Honduras	77	0.45	0.15	4.59	1784.7	4.5	850	Low Trails
26	Hungary	3189	0.1	0.15	5.94	1163	4.3	16780	Moderate Trails
27	Iceland	652	0.2	0.4	5.07	1159.3	4.4	90741	Moderate Trails
28	Japan	5833	0.48	0.04	5.545	1840.25	4.2	62140	High Trails
29	Kazakhstan	25	0.65	0.05	6.515	2224.8	4.5	734	Low Trails
30	Kenya	20	0.45	0.35	8.95	3220.6	4.6	1146	Low Trails
31	Liechtenstein	315	0.5	0.1	5.79	1585.8	4.2	1919	Low Trails
32	Madagascar	23	0.4	0.15	15.93	3100.8	4.6	121	Low Trails
33	Malaysia	4986	0.25	0	2.53	1090.1	4.3	194055	Moderate Trails
34	Mongolia	1	1	0	7	2572	4.5	52	Low Trails
35	Montenegro	80	0.7	0.05	7.03	2318.7	4.4	6113	Low Trails
36	Morocco	24	0.15	0.3	6.23	2342.3	4.3	2470	Low Trails
37	Namibia	33	0.05	0.35	5.02	1026.6	4.5	174	Low Trails
38	Nepal	67	0.9	0.05	26.6	12385.9	4.4	7691	Low Trails
39	New Zealand	5627	0.6	0.05	6.41	2504.1	4.3	358492	High Trails
40	Norway	9124	0.5	0.05	5.94	2128.7	4.2	168672	High Trails
41	Peru	2012	0.9	0	6.36	2195.3	4.5	26877	Moderate Trails
42	Philippines	1891	0.6	0.15	3.86	1274.2	4.2	5085	Moderate Trails
43	Portugal	9433	0.2	0.1	5.08	1845.5	4.1	258188	High Trails
44	Romania	1175	0.6	0.05	5.86	2367.4	4.4	7485	Moderate Trails
45	Slovenia	657	0.25	0.2	4.04	967.1	4.3	24756	Moderate Trails
46	Spain	23696	0.55	0	6.3	1715.5	4.1	455144	High Trails
47	Sweden	8283	0	0.15	5.46	718.7	4.1	112338	High Trails
48	Switzerland	16413	0.3	0.12	4.42	1046.222222	4.3	243491	High Trails
49	Tanzania	21	0.8	0.05	177.5	6068.1	4.8	2016	Low Trails
50	Thailand	2537	0.38	0.04	4.13	7964.95	4	30018	Moderate Trails
51	Turkey	120	0.3	0.15	7.02	1339.1	4.3	13503	Low Trails
52	United Kingdom	25615	0.2	0	7.96	1930.7	4.3	2238887	High Trails
53	United States	112912	0.68	0.02	5.144	1458.15	4.3	34882746	High Trails
54	Vietnam	160	0.25	0.1	4.108	1567.1	4.3	9684	Low Trails

Prepping for Decision Tree Model:

To start off, we need a label column for this data set. Our TrailLevel column label bins the TrailCount variable based on the amount of trails in the country. This provides us with a meaningful target for our supervised learning. So “Low Trails” are locations that do not have many hiking trails, while “high trails” are areas with a high count of trails in the area, with “moderate trails” being a number between.

The 'Country' column must be removed because it does not directly affect our classification so do not believe it will add any meaningful insight. The data is already cleaned from prepping for KMeans in a different module, so after Country has been dropped, we can split our data into training and testing.

Format the Labeled Data you plan to use with Decision Tree Modeling

Below I have split the testing and training data, and preserved the labels before dropping them.

The screenshot displays a Jupyter Notebook interface with a code editor on the left and a console/output area on the right. The code defines a decision tree model for traffic classification based on the 2015 Kaggle Survey of Internet Usage dataset.

Code Snippets:

```

import pandas as pd
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import train_test_split
import os
from sklearn.metrics import confusion_matrix
from sklearn.metrics import classification_report

# Load data
filename = "data/traffic_classification.csv"
train_data = pd.read_csv(filename)

# Drop Country column
train_data.drop('Country', inplace=True)

# Split data
train_data, test_data = train_test_split(train_data, test_size=0.2, random_state=1)

# Train model
model = DecisionTreeClassifier()
model.fit(train_data, train_data['label'])

# Evaluate model
train_data['label'] = model.predict(train_data)
test_data['label'] = model.predict(test_data)

# Confusion matrix and classification report
cm = confusion_matrix(test_data['label'], model.predict(test_data))
print(classification_report(test_data['label'], model.predict(test_data)))

```

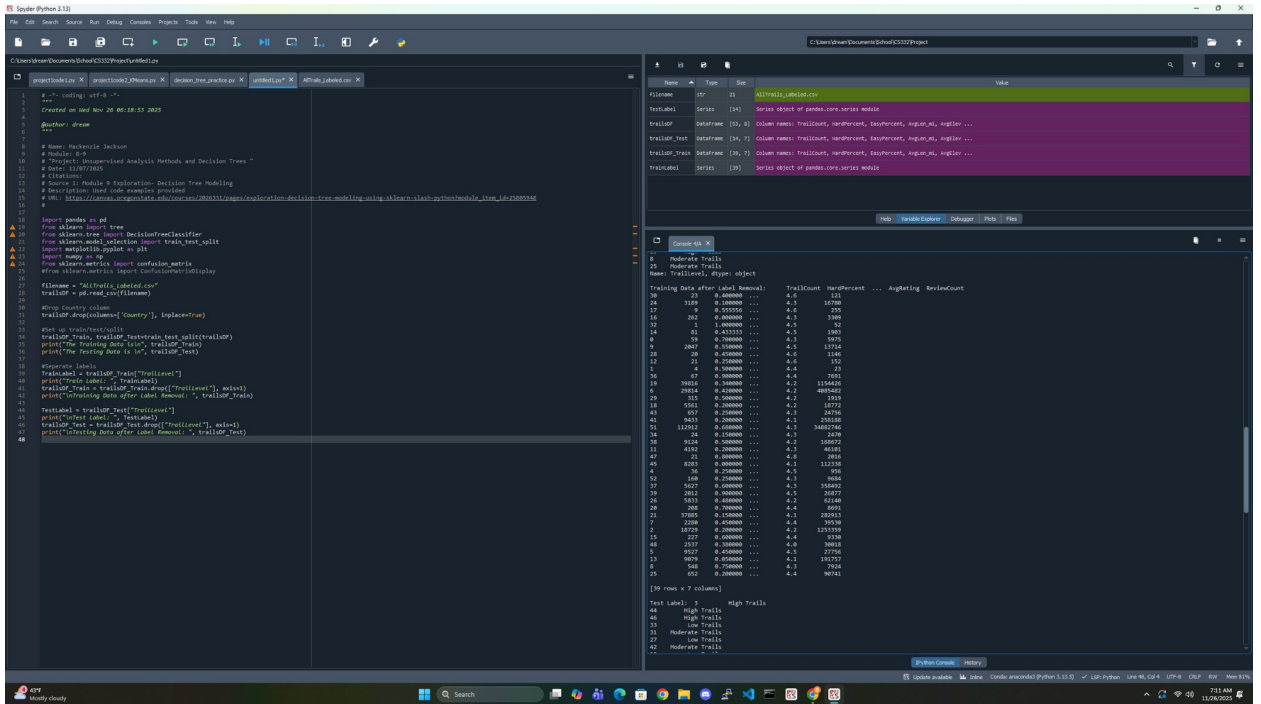
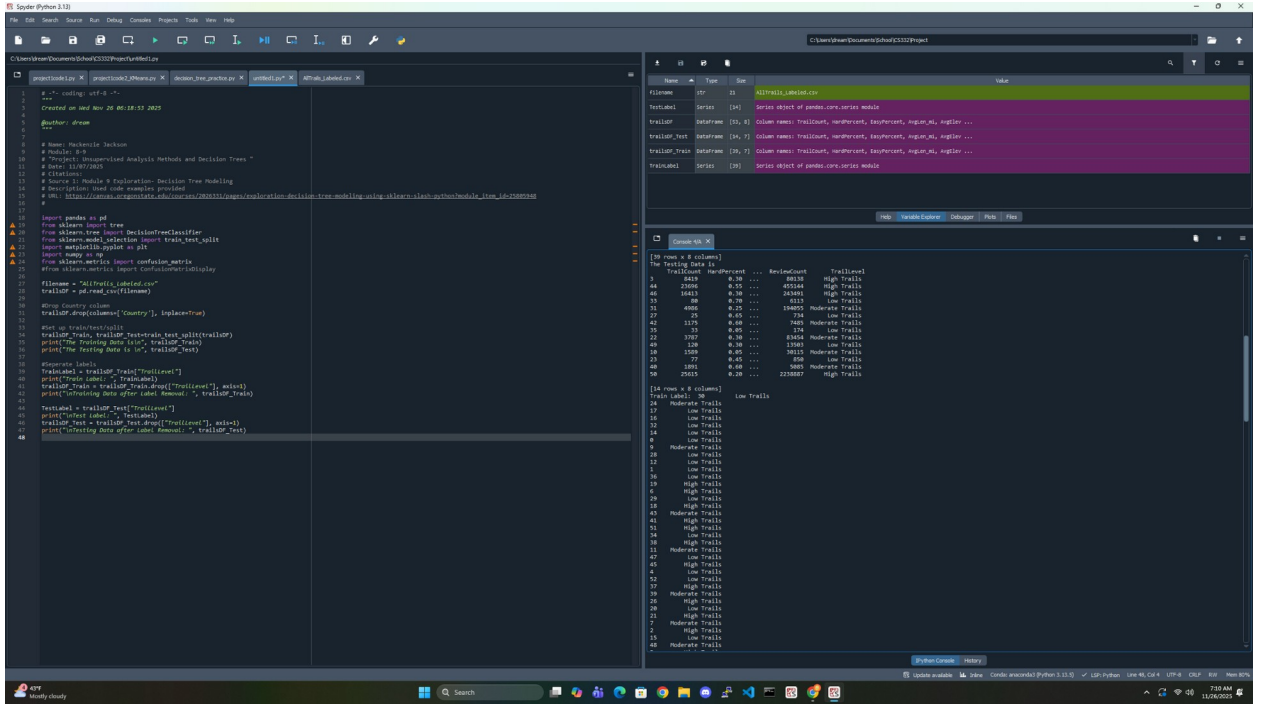
Output:

```

Python 3.5.3 | packaged by Anaconda, Inc. | (env: base) Jan 12, 2018 16:37:07 [MC V.1929.64 bit (64064)]
Type "help()" for credits or "license()" for more information.
Python 3.5.3 > An enhanced interactive Python. Type '?' for help.
Python 3.5.3 > C:\Users\jason\Documents\SurveyOfInternetUsage\2015\Project\traffic_test.py --help
The Training Data Is:
  TrainCount  nonPercent ...  TrainCount  TrainLabel
24  23  0.000000 ...  221  Low Train
25  1109  0.000000 ...  1078  Moderate Train
26  8  0.000000 ...  205  Low Train
27  262  0.000000 ...  1001  Low Train
28  2  0.000000 ...  32  Low Train
29  81  0.433333 ...  1097  Low Train
30  79  0.000000 ...  323  Low Train
31  2047  0.000000 ...  1514  Moderate Train
32  20  0.000000 ...  1145  Low Train
33  11  0.000000 ...  132  Low Train
34  6  0.000000 ...  73  Low Train
35  67  0.000000 ...  7651  Low Train
36  78856  0.000000 ...  118463  High Train
37  28  0.000000 ...  1553  Low Train
38  315  0.000000 ...  2772  High Train
39  657  0.000000 ...  2476  Moderate Train
40  8413  0.000000 ...  20158  High Train
41  112712  0.000000 ...  1482124  High Train
42  24  0.000000 ...  276  Low Train
43  9124  0.000000 ...  14687  High Train
44  4192  0.000000 ...  40168  Moderate Train
45  21  0.000000 ...  2851  Low Train
46  8293  0.000000 ...  11737  High Train
47  126  0.000000 ...  956  Low Train
48  2667  0.000000 ...  30883  High Train
49  2812  0.000000 ...  26877  Moderate Train
50  9413  0.000000 ...  41244  High Train
51  788  0.000000 ...  8051  Low Train
52  17045  0.000000 ...  28113  High Train
53  2208  0.000000 ...  129103  Moderate Train
54  18729  0.000000 ...  129103  High Train
55  227  0.000000 ...  9338  Low Train
56  2537  0.000000 ...  29812  Moderate Train
57  9522  0.000000 ...  27770  High Train
58  929  0.000000 ...  107270  High Train
59  508  0.000000 ...  7028  Moderate Train
60  852  0.000000 ...  98741  Moderate Train

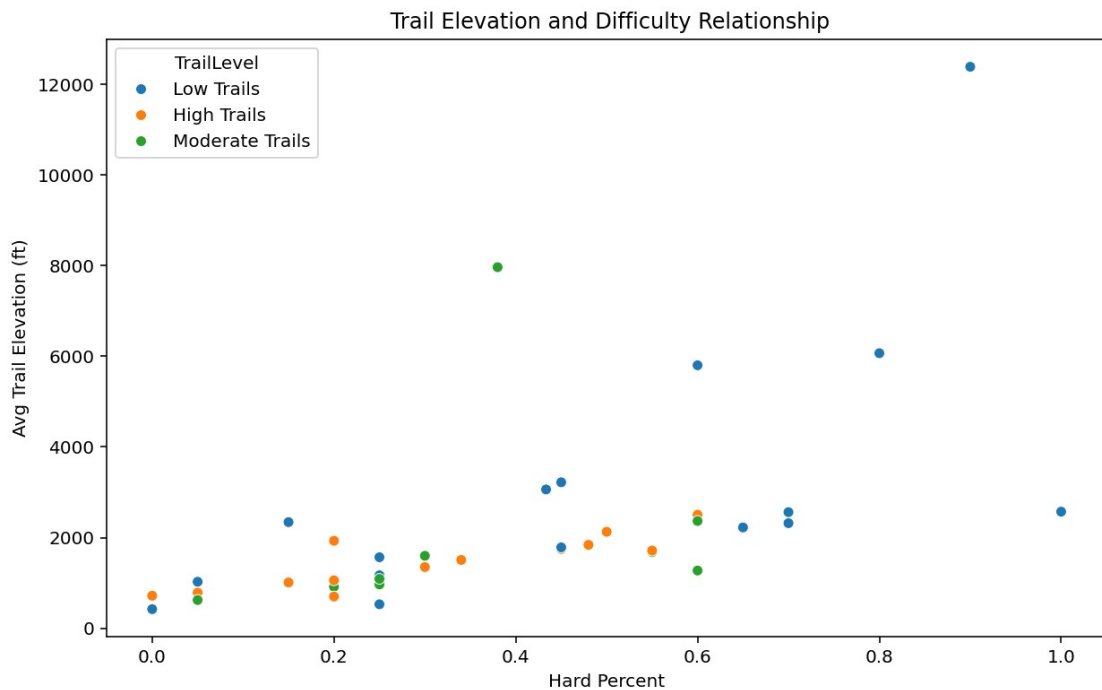
The Test Data Is:
  TrainCount  nonPercent ...  TrainCount  TrainLabel
41  8413  0.39 ...  88138  High Train
42  23066  0.50 ...  435444  High Train
43  14415  0.76 ...  304651  High Train

```



This boxplot tells how hard (difficulty) percent varies across the three labeled groups. Countries with a low number of trails have the widest spread. This suggests that the low-trail countries can either be dominated by easier routes or by more difficult ones, or in between. The moderate-trail count group also spans broadly but the central box is slightly more narrow and lower, which shows that there is a more balanced distribution with difficulty. The high trail group has the most narrow distribution and does not reach any extreme high hard percents. Overall, it seems with this pattern that as the number of trails increase, the overall difficulty mix leans more toward moderate and stable, while countries with fewer trails are more likely to skew toward either very easy or very difficult trails. Based on this distribution, it appears that countries with a moderate to high number of trails tend to offer a larger share of moderate-difficulty over hard ones. .

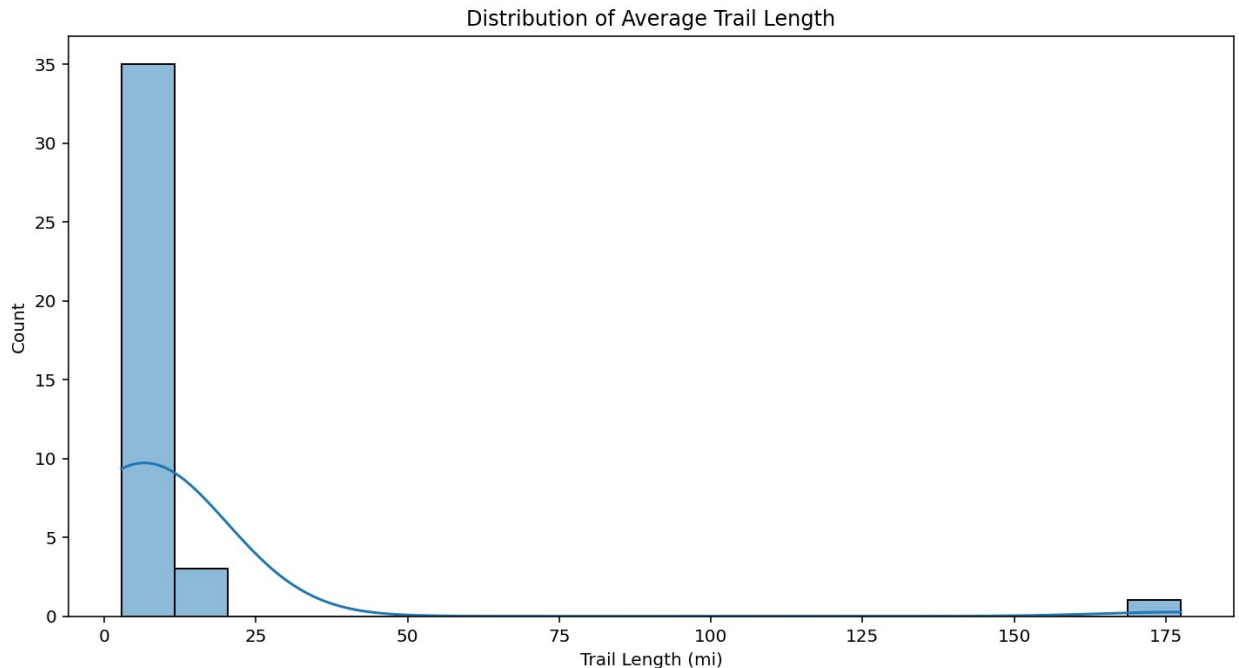
Traillength vs Difficulty:



This scatterplot shows how the average trail elevation gain and the difficulty percentage interact across the three labels. Countries with a large number of trails are clustered more into less difficult trails with lower elevation gain, which likely suggests more accessible trails. As the average trail elevation rises and average hard percentages increase, we see more areas with lower amounts of trails. There are also less countries represented as we increase both variables, suggesting that there aren't many countries that offer more physically demanding and steep trails. Areas with moderate amounts of trails mostly follow the trend of less to moderate hard percent trails and less trail elevation gain. One clear pattern I do see, is that with the increase of average hard percents there is an increase of elevation gain. That is to say, areas with a higher average of hard difficulty trails also have a higher average of elevation gains. This

suggests that elevation gain may be a strong factor in influencing how trails are classified by difficulty.

Distribution of Average Trail Length:



The histogram of average trail length shows a clear skew toward trails with shorter hiking distances. The majority of countries have an average trail length under 10 miles, which indicates that globally, hiking systems tend to prioritize those shorter to moderate trail routes that are likely more accessible, rather than longer distance hiking. A small number of destinations fall into the 10-10 mile range. Beyond that, the distribution drops very sharply with only a few extreme outliers that represent locations with unusually long average trail lengths. This likely correlates to mountainous, more rough terrain, regions. Overall, this pattern suggests that most hiking regions are structured around more shorter/recreational routes suitable for a range of hikers while very long average trail lengths are much more uncommon and may point to more adventure type environments.

Decision Tree Model

In order to set my code up to build the decision-tree model, I instantiated a decision tree classifier using scikit-learn. This created my own object that I can use to train my cleaned and labeled alltrails data. I then fit the model with the training portion of the dataset along with the training labels. Afterwards, I created my prediction for the testing dataset:

```

#plt.xlabel("Hard Percent")
#plt.ylabel("Avg Trail Elevation (ft)")
#plt.show()

#####

#Instantiate/fit model
MyDT_Classifier = DecisionTreeClassifier()
MyDT_Classifier = MyDT_Classifier.fit(trailsDF_Train, TrainLabel)
print("\nDecision Tree Classes: ", MyDT_Classifier.classes_)
print("\nMy Training Columns: ", trailsDF_Train.columns)

#Predict the Testing Dataset
Prediction=MyDT_Classifier.predict(trailsDF_Test)
print("\nMy Prediction:", Prediction)

```

```

[14 rows x 7 columns]

Decision Tree Classes: ['High Trails' 'Low Trails' 'Moderate Trails']

My Training Columns: Index(['TrailCount', 'HardPercent', 'EasyPercent',
                             'AvgLen_mi',
                             'AvgElevGain_ft', 'AvgRating', 'ReviewCount'],
                             dtype='object')

My Prediction: ['Low Trails' 'Low Trails' 'High Trails' 'Low Trails' 'Moderate
Trails'
'Moderate Trails' 'Low Trails' 'High Trails' 'Low Trails' 'High Trails'
'High Trails' 'Low Trails' 'Low Trails' 'High Trails']

In [12]:

```

Next, I created the actual confusion matrix visualization:

```

#Create Confusion Matrix
con_matrix = confusion_matrix(Actual_Labels, Predicted_Labels)
print("\nConfusion Matrix Data: ", con_matrix)

sns.heatmap(
    con_matrix,
    annot=True,
    cmap='Greens',
    xticklabels=label_names,
    yticklabels=label_names,
    cbar=False
)
plt.title("Confusion Matrix for Trail Count Density")
plt.xlabel("Actual")
plt.ylabel("Predicted")
plt.show()

```

```

Decision Tree Classes: ['High Trails' 'Low Trails' 'Moderate Trails']

My Training Columns: Index(['TrailCount', 'HardPercent', 'EasyPercent',
                             'AvgLen_mi',
                             'AvgElevGain_ft', 'AvgRating', 'ReviewCount'],
                             dtype='object')

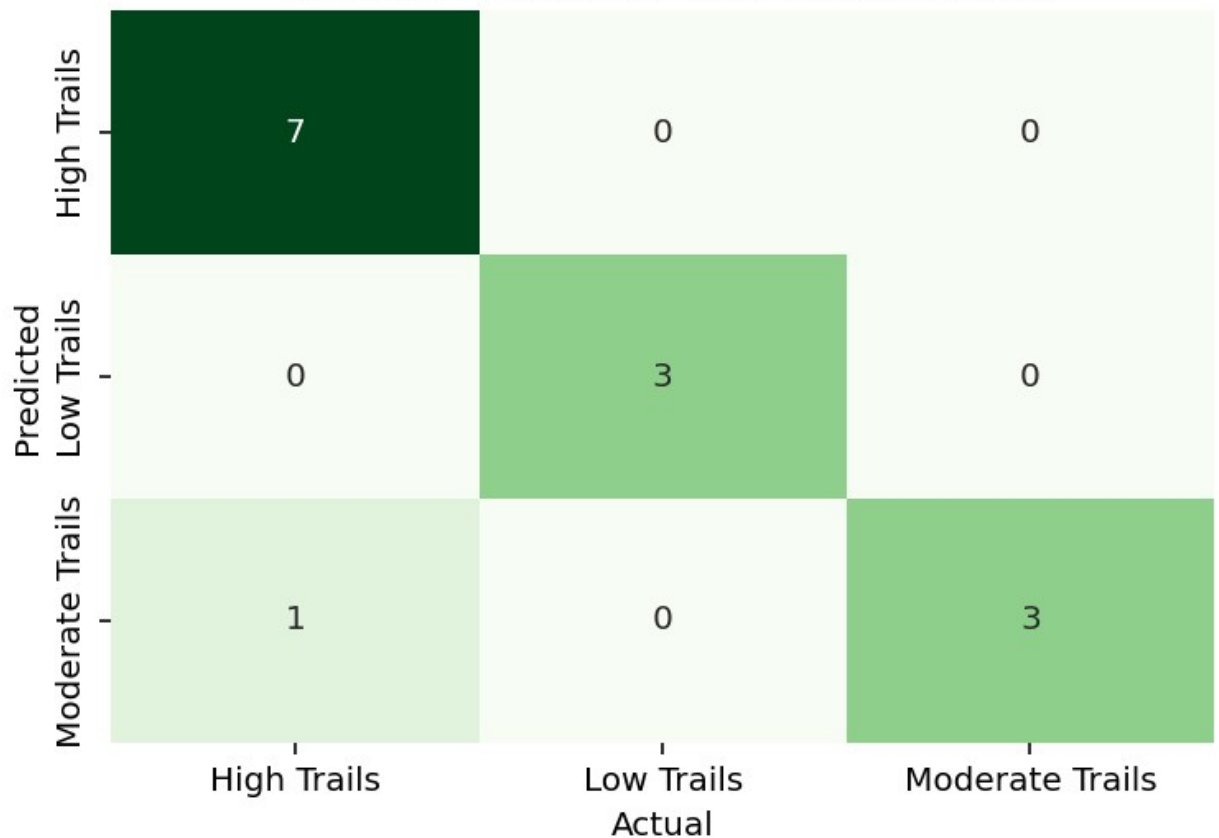
My Prediction: ['High Trails' 'Moderate Trails' 'High Trails' 'Low Trails'
'Moderate Trails' 'Low Trails' 'High Trails' 'Low Trails' 'High Trails'
'High Trails' 'High Trails' 'High Trails' 'High Trails' 'Moderate Trails']

Confusion Matrix Data: [[7 0 0]
 [0 3 0]
 [1 0 3]]

In [14]:

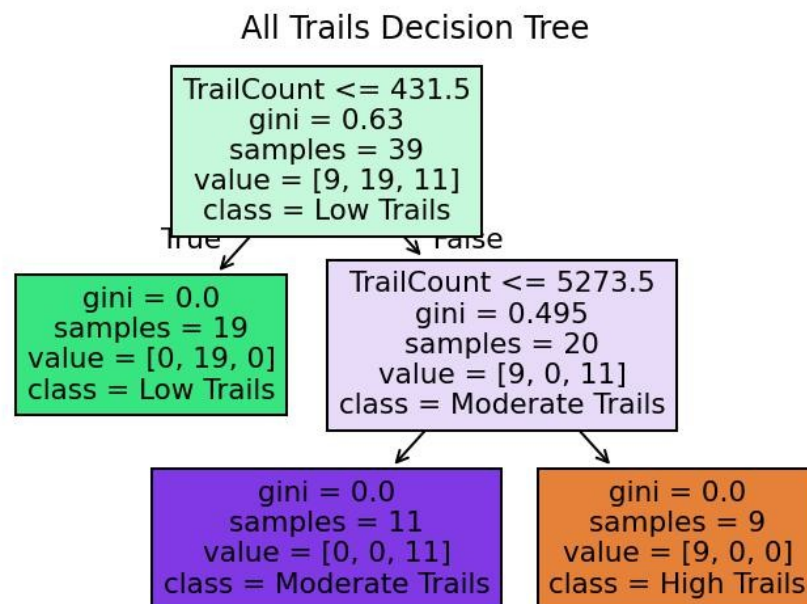
```

Confusion Matrix for Trail Count Density



The confusion matrix I generated shows that the decision tree model performed well overall in predicting the trail category labels. It correctly identified all the “High Trail” and “Low Trail” examples that were in the testing data set. There was one small miscalculation that occurred in the “Moderate Trails” category, when it was mislabeled as a high trail count. It is likely this occurred because there might be some shared characteristics between the two groups making it a little more ambiguous. The one piece of data it was testing might have been “on the cusp” so to speak. Overall, the results showed a strong prediction outcome where the decision tree was mostly successful in determining the amount of trails that were likely for a given destination.

Decision Tree Visualization:



Summary

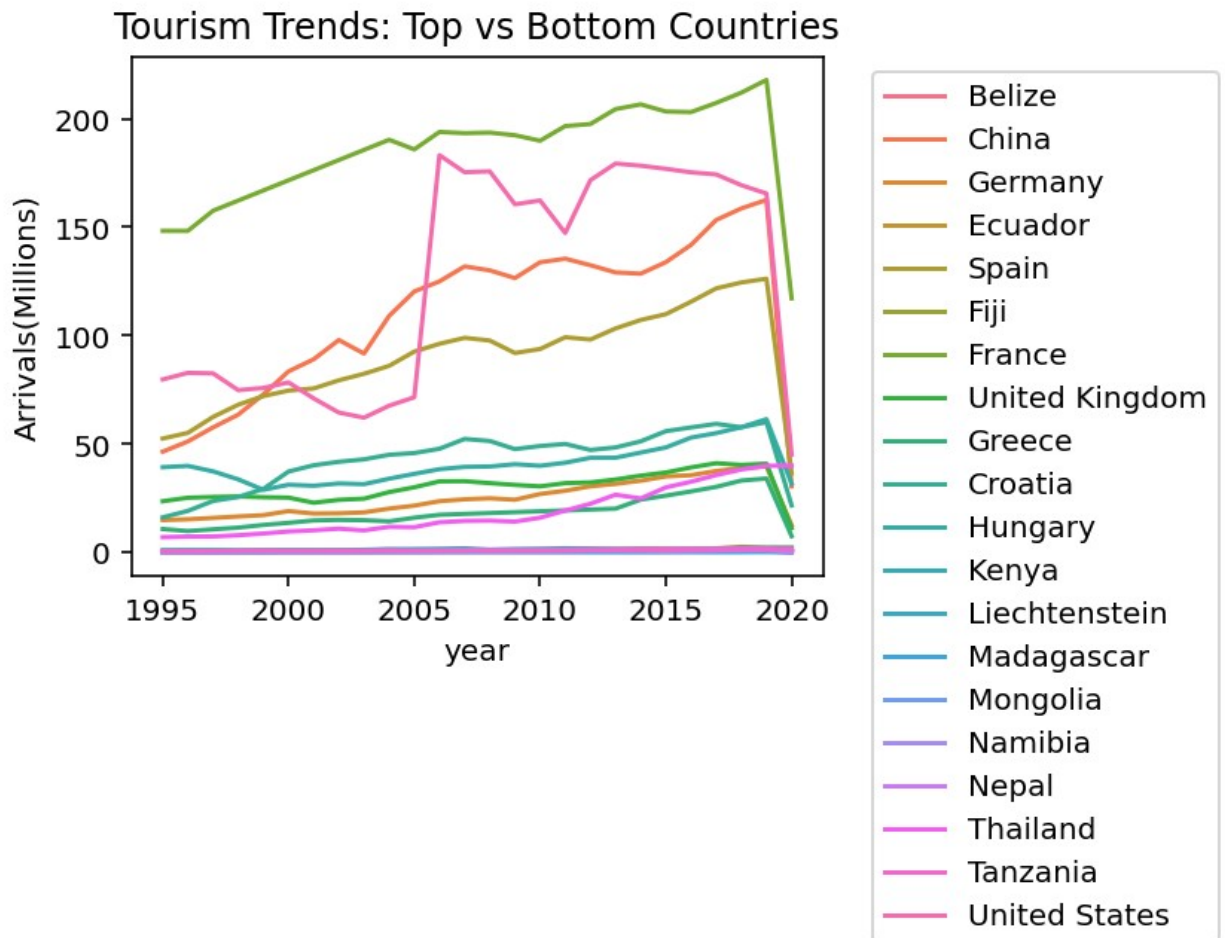
The results from the decision tree model helped make sense of how different trail systems around the world compare. It identified patterns in the data and grouped countries into three broad categories: those with low trail density, moderate trail density, and high trail density. One of the most important discoveries is that countries with a high number of trails don't necessarily have more difficult or longer trails. In a lot of cases, places that had many trails tended to have shorter and easier routes overall, which matches the visualizations earlier in the project. It is safe to say that this points to how accessibility, infrastructure, and tourism likely play a strong role in how large trail infrastructure is formed in those areas, instead of just being based on geography or difficulty of the terrain. This model performed well and was almost completely accurate in its predictions, with only one misclassification.

Additional Analysis

Outside of the required assignments, additional steps were taken in merging the three cleaned datasets. Among other things, this allowed for more visualizations that could assist in showing relationships between tourism numbers, climate patterns, as well as trail characteristics. For example:

- Do tourism arrivals increase or decline over time with certain countries?
- Are extreme temperatures related to lower or possibly fluctuating tourism patterns?
- Does precipitation (rainfall) or seasonal weather variation appear to influence tourism demand?

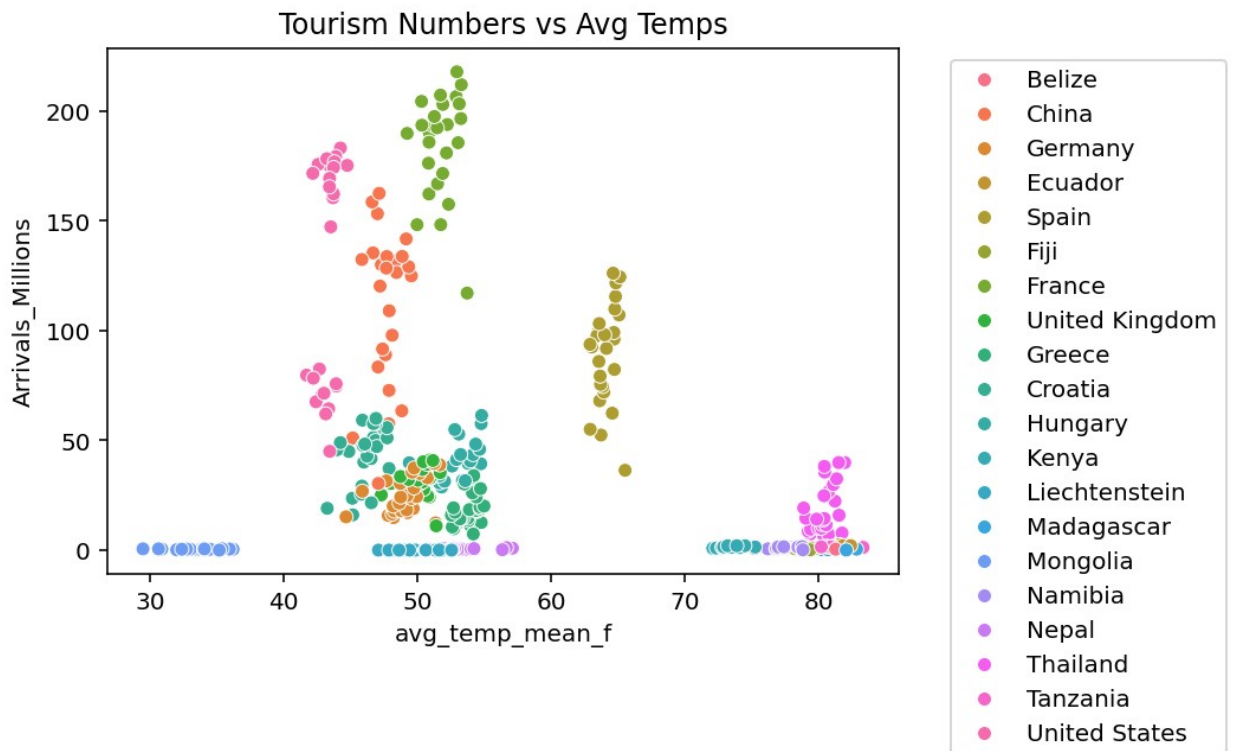
Tourism Trends:



In the above graph, the top and bottom ten countries (arrival numbers) were filtered in order to focus on two extreme groups. There are clear differences between high tourism and low tourism countries. Countries that were highly visited tend to maintain steady or gradually increasing tourism numbers over time, with or without fluctuations. These destinations that have more established tourism are less sensitive to short term changes

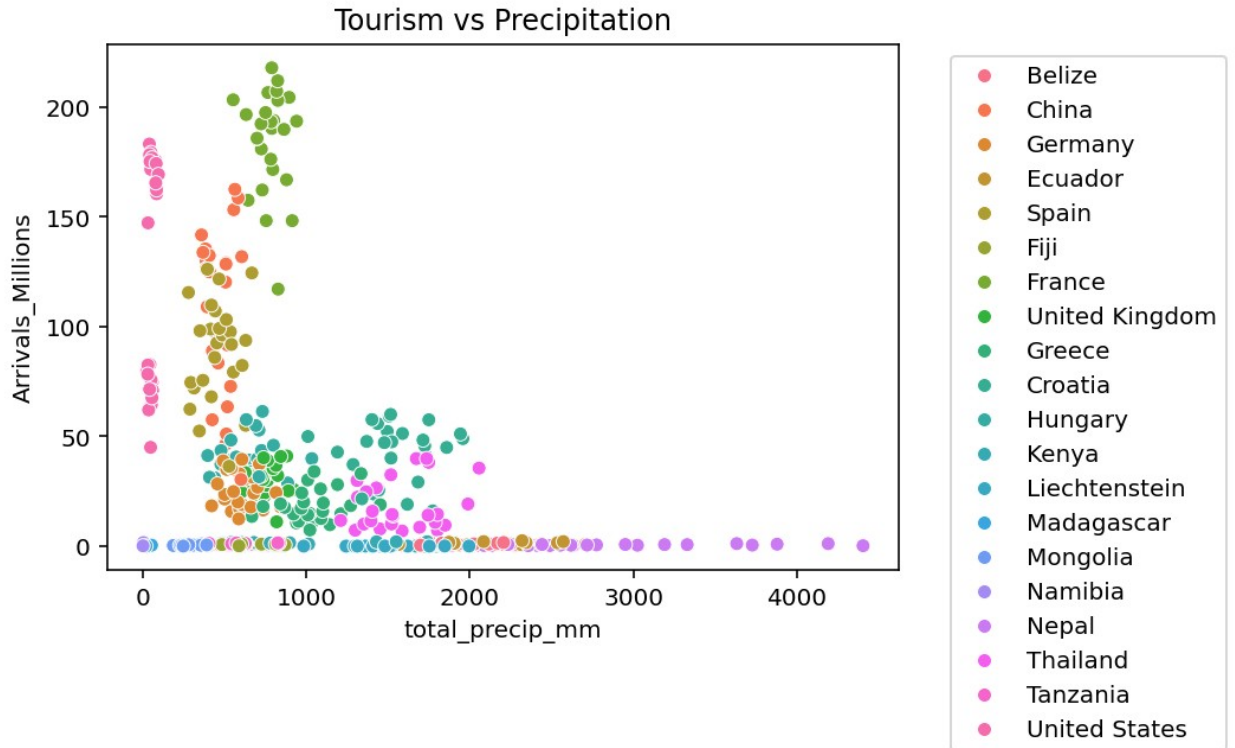
that are likely due from economic shifts or climate anomalies. (For instance, we see a sharp decline in The United States right after the attack of 9/11/2001 that recovered by 2005, as well as a decline around the 2008 recession.) Less visited destinations still saw an increase of tourism on a smaller scale, with nearly all countries showing a sharp decline in 2020, due to the global pandemic.

Tourism Numbers vs Average Temperatures:



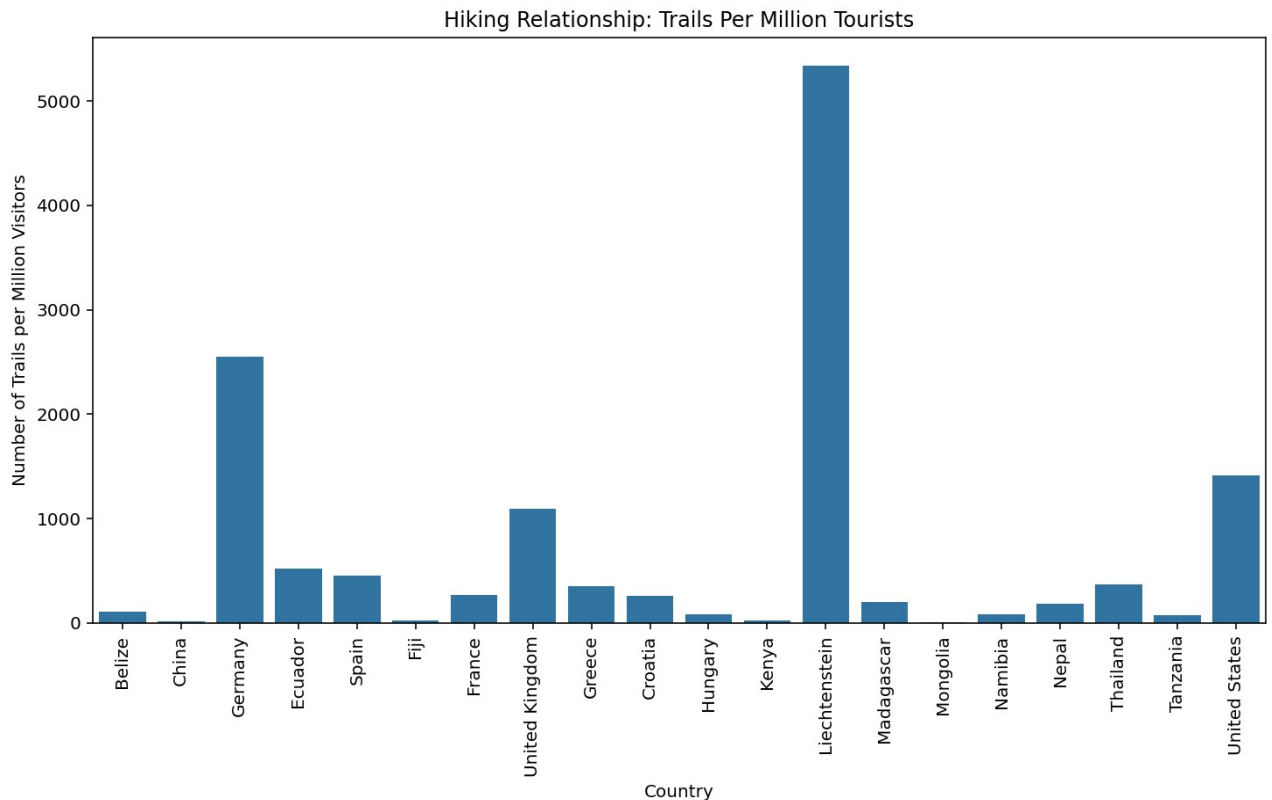
This plot compares mean annual temperatures against the total tourist arrivals for all focus countries. There is no linear relationship that appears between the average temperatures and tourism levels. However, countries with more moderate climates tend to cluster toward higher tourist arrivals. This suggests that extremely hot or extremely cold places may be less appealing to larger scale tourism. There are some outliers, such as Thailand, that still attract a good number of tourists despite hot temperatures. These results indicate that temperature alone does not determine tourism demand, but moderate climates is likely a factor for high visitation numbers.

Tourism Numbers vs Total Precipitation:



This scatterplot shows the relationship between average precipitation and tourism arrival counts across the focus countries. There is a visible pattern: countries with extremely high rainfall fall into lower tourism groups. On the other hand, many destinations that see a lot of tourism cluster around more moderate precipitation levels. Heavy rainfall may shorten the ideal hiking seasons or reduce accessibility. It could deter certain types of outdoor recreation. However, there lacks a very strong correlation that suggests that tourism industries can adapt. Thailand, for example, has high rainfall yet remains a popular destination. This may indicate that tourists avoid the rainy season rather than avoiding the destination entirely.

Trails per Tourist:



This last visualization compares the number of hiking trails relative to the number of incoming tourists by looking at how many trails exist per one million visitors. This helps distinguish countries where hiking is a more defining pull from countries where it is simply one of many tourism options. Countries with the highest bars appear to offer a large volume of hiking opportunities relative to the number of tourists they receive. Some countries (such as Liechtenstein or Germany), suggest a strong geographic association with outdoor or mountain tourism. Others, like the US or UK, likely show a much more extensive trail infrastructure rather than tourism driven primarily by hiking. It's important to take into account how large certain countries are, such as the U.S., and if it were possible it would be much more helpful to focus on specific regions.

Project Conclusion



Hiking tourism is growing worldwide and has become increasingly more advertised to a range of groups. Trail accessibility shapes where people choose to travel and climate patterns also affect tourism numbers. Understanding such patterns can help travelers choose destinations that match their skill level, avoid seasons with heavy rainfall, wildfire risk, or extreme heat. They can also plan trips based on trail availability and popularity, especially in countries where hiking drives tourism and overcrowding is common. Additionally, this allows travelers to identify emerging hiking destinations that offer those beautiful landscapes but likely have fewer crowds. Local governments can also benefit from looking at patterns, as they can plan infrastructure and staffing around peak tourism seasons and maintain trail systems.

Patterns have shown that tourism has generally increased over time, with only major global events causing drops. Within this increase, some countries attract tourists because of their hiking culture, while other destinations receive large numbers of tourists arrivals, but hiking plays a smaller role. We must also consider that certain countries with a very large amount of tourists, such as the United States, have many different geographical regions. For instance, the western region provides more diverse landscape and trail/national park accessibility and it has been known for their outdoor tourism sectors. There are many people who come to the United States to visit iconic cities such as NYC, that are unrelated to hiking.

There are aspects that influence travel and popular hiking destinations, such as trail difficulty and climate. Climate was found to play a role in travel patterns, though it influences more of when people travel rather than whether they travel at all. There are countries with extreme high rainfall or extreme high/low temperatures that still attract visitors, which suggests that travelers adjust to timing rather than skip destinations. However, there is a pattern between moderate weather and popular destinations, which suggests that those destinations attract more visitors for outdoor activities. Lastly, destinations with a large trail system tended to have more easier to moderate difficulties that were often influenced by elevation and length, rather than very difficult trails likely

with rough terrain and steep inclines. There is a distinct correlation between high tourism, hiking infrastructure, and accessibility.

Based on the results found, entities such as tourism boards and national park systems should invest in maintaining beginner friendly and moderate trails. They should improve accessibility, and create year round options wherever possible. Most travelers want safe, reachable experiences. Countries with high trail density but limited land area can mitigate strain caused from high traffic, such as stronger trail maintenance or implementing visitor caps on popular trails. Emerging hiking destinations can prepare now. Climate change should also be considered in future years, where certain months might become too hot or wet and will shift peak tourist seasons. Overall, these patterns highlight the importance of planning ahead so destinations can continue supporting the hiking sector, without damaging the environments that attract those visitors in the first place.

Source Citations

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